

1 Neon AI Factory Architect Solution Architecture Document

Field	Value
Architecture Revision	REV-000003
Baseline	BASELINE-000001
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Document Status	Machine-generated review draft
Compilation Status	Compiled with documented gaps
Documentation Standard	Neon AI Factory Architect SADM v1.0 candidate
ADC Version	1.0.0-candidate
Generation Status	Generated
Document Maturity	Machine-generated review draft
Professional Gate	Passed
Legacy Substituted	No
Compiler	Architecture Document Compiler v1.0.0
Generated	deterministic

Authority Notice: This document is a **Machine-generated review draft**. Customer-issue eligibility: **Eligible**. Current approval status is established by the accompanying customer issue record. Procurement authority: **No**. Installation authority: **No**.

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- Applicable Requirements Register
- Requirements Traceability Matrix
- Architecture Decision and Approval Record
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2.1 Domain Content Coverage: 87%

This metric means the document has renderable section content. It is not a procurement, compliance, or engineering-readiness claim.

-  **Compute Platform:** Content available
-  **Facilities:** Content available
-  **Network Fabric:** Content available
-  **Operations:** Content unresolved — engineering input required
-  **Procurement Planning:** Content available
-  **Requirements and Compliance:** Content available
-  **Security Architecture:** Content available
-  **Storage Architecture:** Content available
- Procurement ready: **No**

2.2 Architecture-Derived Delivery Summary

This concise summary is derived from the immutable Project Delivery Planning Compiler result. The standalone Project Delivery Plan contains the complete task, dependency, gate, risk and handoff registers.

- Plan maturity: **Indicative**

- Delivery structure: 14 phases, 23 workstreams and 285 tasks
- Governed dependencies: 286
- Risks and engineering handoffs: 0 risks / 71 handoffs
- Indicative completion: Not calculated
- Authority: Architecture-derived planning review only; not a contractual programme, resource-feasibility commitment, procurement authorization or installation authorization.

2.3 Architecture at a Glance

- The selected compute platform is HGX B300.
- The architecture provides 1,152 GPUs.
- The east-west compute fabric uses InfiniBand.
- Plan maturity: **Indicative**

2.4 Network and Connectivity

- Scale-up interconnect uses NVIDIA NVLink / NVSwitch for GPU-to-GPU communication inside each compute domain.
- East-west fabric uses InfiniBand for scale-out workload traffic.
- Storage fabric uses Spectrum-X Ethernet for storage client and system traffic.
- Frontend and north-south fabric uses Ethernet for service and tenant ingress and egress.
- External connectivity: Unresolved — engineering input required
- Out-of-band management fabric uses Ethernet for management, administration and recovery traffic.

2.5 Storage

- WEKA Core AI is represented as a governed architecture review candidate.
- Calculated usable capacity is 4640.91 TB.
- The planning configuration contains 48 WEKA backend nodes.
- WEKA storage throughput: Pending storage engineering / vendor sizing
- Requested usable storage capacity is 4.0 PiB.
- Storage-system power: Not yet calculable — storage configuration required
- Storage fabric demand: Not yet calculable — storage performance and topology inputs required

2.6 Security

- Security architecture guidance includes 2 governed security views.
- Security findings and implementation actions require human review and do not constitute certification or control-effectiveness evidence.

2.7 Architecture Alternative Considered

2 material architecture differences identified.

		Lintasarta 1,024 GPU Physical governed alternative	Main Architecture
#	Parameter		
1	Requested usable storage capacity	4.5 PiB	4 PiB
2	Storage model	High-performance storage with context-memory storage	High-performance storage

2.8 Indicative Architecture Bill of Materials

Indicative architecture BoM — not approved for procurement or installation.

Component	Product / Class	Quantity	Basis	Status
Ethernet storage fabric switch family candidate	SN5610 / SN5600D family	Unresolved	Governed planning basis	Unresolved
External OOB aggregation switch	SN2201 family	Unresolved	Governed planning basis	Unresolved
HGX B300 certified OEM system candidate	HGX B300 8-GPU reference platform	144	Governed planning basis	Resolved
InfiniBand leaf switch candidate	Q3400-RA	16	Governed planning basis	Resolved
InfiniBand spine switch candidate	Q3400-RA	8	Governed planning basis	Resolved
Planning group for management software, support entitlements, observability, lifecycle	Vendor confirmation required	Unresolved	Governed planning basis	Unresolved

Component	Product / Class	Quantity	Basis	Status
tooling and vendor services.				
Planning group for selected core AI storage vendor architecture. Final appliance family, node count, drive class and capacity are vendor-sizing dependent.	Vendor confirmation required	Unresolved	Governed planning basis	Unresolved
Planning group for Spectrum-X host network adapters where the CKP template sources a NIC model.	Vendor confirmation required	Unresolved	Governed planning basis	Unresolved
Planning group for Spectrum-X leaf/spine switch families across requested east-west or storage-fabric roles.	Vendor confirmation required	Unresolved	Governed planning basis	Unresolved
Planning group for storage fabric attachment between compute clients and the selected storage system.	Vendor confirmation required	Unresolved	Governed planning basis	Unresolved

2.9 Essential Diagram Commentary

2.9.1 Overall architecture

- The selected compute platform is HGX B300.
- The architecture provides 1,152 GPUs.
- The east-west compute fabric uses InfiniBand.

2.9.2 Principal network

- Scale-up interconnect uses NVIDIA NVLink / NVSwitch for GPU-to-GPU communication inside each compute domain.
- East-west fabric uses InfiniBand for scale-out workload traffic.
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2.9.4 Security architecture

- Security architecture guidance includes 2 governed security views.
- Security findings and implementation actions require human review and do not constitute certification or control-effectiveness evidence.

2.10 Review Identity

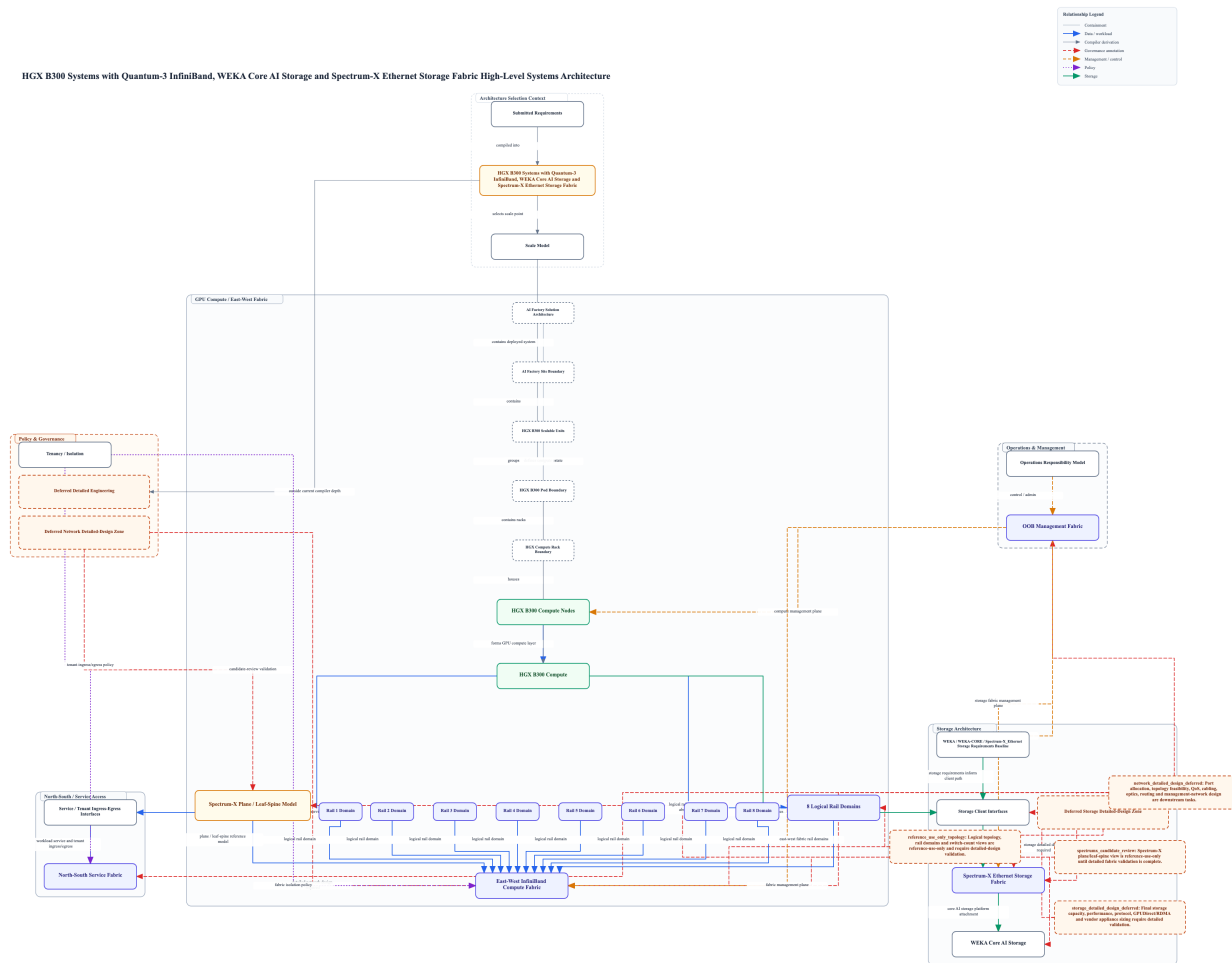
- **Recorded by:** BD Independent SME Reviewer
- **Role:** Architecture approver
- **Recorded at:** 2026-09-01T10:53:43Z

2.11 Main Architecture Visualisation

Section status:  Accepted | Completeness: Unresolved — engineering input required

(No content available for this section.)

2.12 Figure 1: Overall AI Factory System Architecture



Neon House Standard overall AI Factory system architecture

Purpose: Revision-bound Neon House Standard view of the current AI Factory system architecture.

How to read this figure: This Overall AI Factory System Architecture presents its governed architecture domains and explicitly labelled relationships.

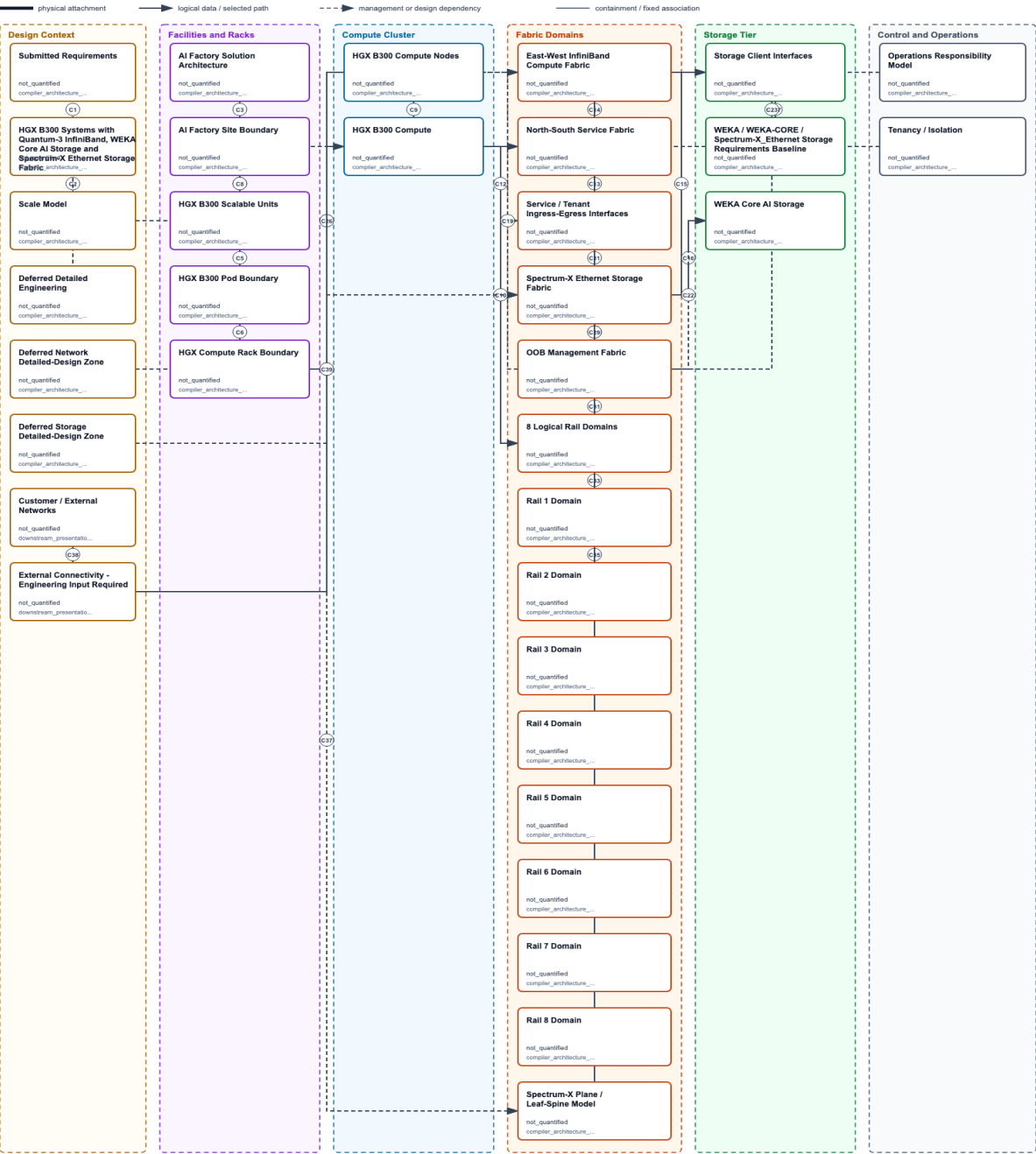
Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.13 Figure 2: Conceptual Physical Topology

Conceptual Physical Topology

Candidate visual projection. Quantities, media and authority are copied from compiler artefacts or shown unresolved.

source_run_id=RUN-DETERMINISTIC-655a35e | diagram_set_id=DS-ET-RUN-DETERMINISTIC-655a35e-CONCEPTUAL-TOPOLOGY-V1 | maturity=Production_Grade_Solution_Architecture | generated_at=2026-09-01T10:52:48Z



Connection schedule

- C1 design_dependency: compiled into
- C2 design_dependency: selects scale point
- C3 containment: contains deployed system
- C4 containment: contains

Conceptual Physical Topology

Purpose: Conceptual Physical Topology

How to read this figure: This Conceptual Physical Topology presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.14 Design Basis

Section status:  Accepted | Completeness: Unresolved — engineering input required

Concept	Value	Detail Level
Architecture selection	HGX B300 Systems with Quantum-3 InfiniBand, WEKA Core AI Storage and Spectrum-X Ethernet Storage Fabric (Architecture-generation authority: No; Architecture-generation status: Not activated; separate governed action required; East-west fabric: InfiniBand; Engineering-handoff authority: No)	L2

2.15 Overall Conceptual Architecture

Section status:  Accepted | Completeness: Unresolved — engineering input required

Concept	Value	Detail Level
Architecture selection	HGX B300 Systems with Quantum-3 InfiniBand, WEKA Core AI Storage and Spectrum-X Ethernet Storage Fabric (Architecture-generation authority: No; Architecture-generation status: Not activated; separate governed action required; East-west fabric: InfiniBand; Engineering-handoff authority: No)	L2

2.16 Compute Platform

Section status:  Candidate — review required | Completeness: Unresolved — engineering input required

The architecture uses HGX B300 Systems with Quantum-3 InfiniBand, WEKA Core AI Storage and Spectrum-X Ethernet Storage Fabric (Architecture-generation authority: No; Architecture-generation status: Not activated; separate governed action required; East-west fabric: InfiniBand; Engineering-handoff authority: No) with 1,024 GPUs. The platform was selected by the governed architecture assessment using the requested GPU count and fabric requirements. The compute platform requires detailed engineering design before procurement or installation.

Key planning facts:

- GPU count: 1,024
- Rack count: Unresolved — engineering input required
- Node count: Unresolved — engineering input required
- Platform: HGX B300 Systems with Quantum-3 InfiniBand, WEKA Core AI Storage and Spectrum-X Ethernet Storage Fabric (Architecture-generation authority: No; Architecture-generation status: Not activated; separate governed action required; East-west fabric: InfiniBand; Engineering-handoff authority: No)

2.17 NVLink/NVSwitch Scale-Up Fabric

Section status:  Accepted | Completeness: Unresolved — engineering input required

The selected platform uses **NVIDIA NVLink / NVSwitch** as its scale-up accelerator fabric — the GPU peer-memory-semantic fabric that ties GPUs together within a single tightly coupled compute domain. This is architecturally distinct from the scale-out compute fabric described in the Network Fabric section: the scale-up fabric operates *within* one domain (one rack for a rack-scale platform, one node for a node-scale platform); the scale-out fabric operates *between* domains.

Key planning facts:

- Domain scope: node_scale
- Switch architecture: baseboard_integrated
- GPUs per scale-up domain: 8
- Scale-up domains in this architecture: 144 (exact)
- NVLink switch trays per domain: Not applicable for this platform
- Compute trays per domain: Not applicable for this platform
- NVLink generation: 5
- Per-GPU GPU-to-GPU bandwidth: 1.8
- Aggregate domain bandwidth: 14.4

These quantities are derived from the selected platform and requested GPU scale — they are not independently architect-configurable for this reference platform. Internal switch wiring, cable schedule and exact port mapping are **not authoritative for procurement or installation** and require detailed design.

2.18 Security Architecture

Section status:  Accepted | Completeness: Unresolved — engineering input required

The baseline security architecture is projected from the selected architecture zones and governed crossings. Architecture-level security zones, governed flows, constraints and findings were compiled for this revision. The result defines architecture-level boundaries and review actions; it is not compliance certification, control-effectiveness evidence, risk acceptance, or deployable configuration.

Advisory scope

- Applied technology overlays: 0
- Architecture/security findings: 26
- System guardrails: 4
- Selected baseline zones: 8
- Governed crossings: 7
- Security architecture guidance patterns: 16 (reference catalogue; not selected topology or deployed enforcement)
- Planning capabilities: 15 (unquantified; not procurement-ready)
- Security Architect handoffs: 25

Architecture/security finding severity

High	Medium	Low
20	5	0

System guardrails

The following 4 critical system guardrails are enforced. They are product boundaries, not critical vulnerabilities in the proposed customer architecture.

- Guardrail 1: Compliance, certification and control effectiveness require separate evidence and review.
- Guardrail 2: Residual risk acceptance remains a human decision.
- Guardrail 3: Architecture guidance does not constitute deployable device, firewall, switch or credential configuration.
- Guardrail 4: Governed architecture facts remain read-only during security assessment.

Baseline security zones — selected architecture

Zone	Architecture purpose	Source authority
zone.compute	Compute Fabric Zone	Selected architecture baseline
zone.external	External / Customer Zone	Selected architecture baseline
zone.management	Management Zone	Selected architecture baseline
zone.oob	OOB Management Zone	Selected architecture baseline
zone.operations	Operations / Logging Zone	Selected architecture baseline
zone.platform	Platform Services Zone	Selected architecture baseline
zone.storage	Storage Zone	Selected architecture baseline
zone.tenant_workload	Tenant Workload Zone	Selected architecture baseline

Governed baseline crossings

Source	Destination	Flow class	Disposition
zone.management	zone.oob	privileged_administration	unresolved_detailed_design
zone.compute	zone.storage	storage_data_path	architecturally_required
zone.external	zone.platform	identity_assertion	unresolved_external_dependency
zone.platform	zone.compute	control_plane	architecturally_required
zone.storage	zone.operations	telemetry	architecturally_required
zone.compute	zone.operations	telemetry	architecturally_required
zone.external	zone.platform	workload_submission	architecturally_required_controls_unresolved

Explicit-allow and deny-unless-required values are security architecture guidance. They do not prove current deployed permission, prohibition, or control effectiveness.

Architecture findings

Finding	Severity	Domain	Required architecture action
Finding 1	High	oob	OOB management must be represented as a distinct restricted security zone separated from

Finding	Severity	Domain	Required architecture action
			user/workload access.
Finding 2	High	oob	Switch management interfaces must be isolated from workload and user access paths.
Finding 3	High	oob	User-plane reachability to IPMI/BMC management services must be denied by architecture.
Finding 4	High	oob	Any OOB route handoff to customer networks must be explicit, bounded, and identified as privileged management exposure.
Finding 5	Medium	oob	The architecture assessment must report whether OOB management has redundant uplinks/paths when upstream facts provide them.
Finding 6	Medium	oob	OOB administrative access and device-management events require a security logging/monitoring handoff.
Finding 7	High	control_plane	Observability backends containing infrastructure telemetry must be placed in a restricted monitoring zone.
Finding 8	High	control_plane	Administrative identity, MFA/PAM and break-glass requirements must be emitted as implementation handoffs; The architecture assessment does not assert effectiveness.
Finding 9	Medium	control_plane	Unknown platform/operator version support must remain unresolved rather than assumed

Finding	Severity	Domain	Required architecture action
Finding 10	High	fabric	compatible. Compute-fabric access must be limited to workload communication and authorized management paths; external/user ingress must not be inferred.
Finding 11	High	fabric	Where upstream design distinguishes storage and compute fabrics, The architecture assessment must preserve those boundaries rather than flattening them.
Finding 12	High	fabric	NVLink/NVSwitch scale-up domain must be represented as a distinct high-trust GPU communication domain and not exposed as routable user network.
Finding 13	High	tenancy	Unknown tenancy mode must prevent automatic selection of single-tenant or multi-tenant security overlays.
Finding 14	High	confidential	GPU confidential-computing capability must remain unknown unless the hardware SKU/capability is evidenced upstream.
Finding 15	High	storage	High-performance storage must be represented as a distinct data/security zone with explicit compute-to-storage flows.
Finding 16	High	storage	Storage administrative interfaces must terminate in privileged management zones and not

Finding	Severity	Domain	Required architecture action
			user/workload zones.
Finding 17	Medium	storage	Unknown encryption-at-rest state must remain unresolved and cannot be reported satisfied.
Finding 18	High	storage	Storage authentication credentials/keys are implementation artifacts and must never be generated by the security architecture assessment.
Finding 19	High	external	Customer edge/uplink connectivity must be represented as an external trust boundary with explicit route/flow classes.
Finding 20	High	external	Unknown external exposure must remain unresolved and visible in the architecture assessment output.
Finding 21	High	observability	Security and infrastructure telemetry stores must be represented as restricted services.
Finding 22	High	observability	Administrative actions require an audit/telemetry handoff; The architecture assessment reports architecture requirement not logging effectiveness.
Finding 23	Critical	identity	Administrative identity roles must be distinct from ordinary user/workload identities.
Finding 24	High	identity	Emergency/break-glass administrative access must be a governed implementation handoff, not auto-generated by the security

Finding	Severity	Domain	Required architecture action
Finding 25	High	identity	architecture assessment. Unknown MFA/PAM state must remain unresolved; platform support does not prove deployment.
Finding 26	Medium	supply_chain	Version-specific NVIDIA reference-architecture support claims require explicit version facts; otherwise remain unresolved.

Required Security Architecture Capabilities — Planning Only

These entries identify capabilities required or recommended by the compiled security architecture. They do not assert that the capabilities are deployed, licensed, configured, effective or quantitatively sized.

- OOB management segmentation — quantity unresolved; procurement and installation authority: No.
- Admin control-plane isolation — quantity unresolved; procurement and installation authority: No.
- User control-plane boundary — quantity unresolved; procurement and installation authority: No.
- BlueField root-of-trust/secure management — quantity unresolved; procurement and installation authority: No.
- BlueField attestation — quantity unresolved; procurement and installation authority: No.
- GPU attestation — quantity unresolved; procurement and installation authority: No.
- Attestation policy/key broker services — quantity unresolved; procurement and installation authority: No.
- Kata confidential sandbox — quantity unresolved; procurement and installation authority: No.
- Compute/management fabric segmentation — quantity unresolved; procurement and installation authority: No.
- Cross-plane tenant isolation — quantity unresolved; procurement and installation authority: No.
- Security observability/audit — quantity unresolved; procurement and installation authority: No.
- Artifact provenance policy — quantity unresolved; procurement and installation authority: No.
- External boundary controls — quantity unresolved; procurement and installation authority: No.
- Storage identity/isolation — quantity unresolved; procurement and installation authority: No.
- Privileged/user/service identity separation — quantity unresolved; procurement and installation authority: No.

Security engineering handoff

- Action 1: OOB management must be represented as a distinct restricted security zone separated from user/workload access.
- Action 2: Switch management interfaces must be isolated from workload and user access paths.
- Action 3: User-plane reachability to IPMI/BMC management services must be denied by architecture.
- Action 4: Any OOB route handoff to customer networks must be explicit, bounded, and identified as privileged management exposure.
- Action 5: OOB administrative access and device-management events require a security logging/monitoring handoff.
- Action 6: Observability backends containing infrastructure telemetry must be placed in a restricted monitoring zone.
- Action 7: Administrative identity, MFA/PAM and break-glass requirements must be emitted as implementation handoffs; The architecture assessment does not assert effectiveness.
- Action 8: Unknown platform/operator version support must remain unresolved rather than assumed compatible.
- Action 9: Compute-fabric access must be limited to workload communication and authorized management paths; external/user ingress must not be inferred.
- Action 10: Where upstream design distinguishes storage and compute fabrics, The architecture assessment must preserve those boundaries rather than flattening them.
- Action 11: NVLink/NVSwitch scale-up domain must be represented as a distinct high-trust GPU communication domain and not exposed as routable user network.
- Action 12: Unknown tenancy mode must prevent automatic selection of single-tenant or multi-tenant security overlays.
- Action 13: GPU confidential-computing capability must remain unknown unless the hardware SKU/capability is evidenced upstream.
- Action 14: High-performance storage must be represented as a distinct data/security zone with explicit compute-to-storage flows.
- Action 15: Storage administrative interfaces must terminate in privileged management zones and not user/workload zones.
- Action 16: Unknown encryption-at-rest state must remain unresolved and cannot be reported satisfied.
- Action 17: Storage authentication credentials/keys are implementation artifacts and must never be generated by the security architecture assessment.
- Action 18: Customer edge/uplink connectivity must be represented as an external trust boundary with explicit route/flow classes.
- Action 19: Unknown external exposure must remain unresolved and visible in the architecture assessment output.
- Action 20: Security and infrastructure telemetry stores must be represented as restricted services.
- Action 21: Administrative actions require an audit/telemetry handoff; The architecture assessment reports architecture requirement not logging effectiveness.
- Action 22: Administrative identity roles must be distinct from ordinary user/workload identities.
- Action 23: Emergency/break-glass administrative access must be a governed implementation handoff, not auto-generated by the security architecture assessment.
- Action 24: Unknown MFA/PAM state must remain unresolved; platform support does not prove deployment.

- Action 25: Version-specific NVIDIA reference-architecture support claims require explicit version facts; otherwise remain unresolved.

2.19 Facilities Envelope

Section status:  Accepted | Completeness: Unresolved — engineering input required

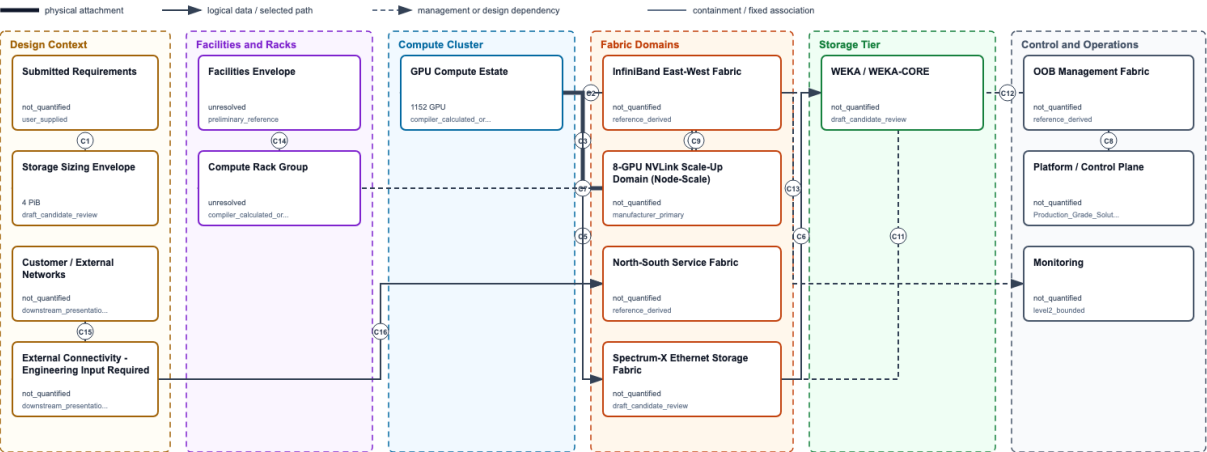
The facilities envelope provides preliminary power, cooling, and site planning information. A facilities envelope has been calculated based on the selected compute and network scale. The facilities assessment remains **planning-only** and requires site-specific engineering before detailed MEP design or procurement.

2.20 Figure 3: Conceptual Logical and Control Flows

Conceptual Logical and Control Flows

Candidate visual projection. Quantities, media and authority are copied from compiler artefacts or shown unresolved.

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Connection schedule

- C1 design_dependency: sizing constraints
- C2 logical_data_flow: collective workload traffic
- C3 physical_attachment: GPU peer-memory-semantic scale-up fabric
- C4 logical_data_flow: External scale-out fabric boundary
- C5 logical_data_flow: storage client I/O
- C6 logical_data_flow: selected storage protocols
- C7 logical_data_flow: service, client and external ingress-egress
- C8 management_control: administration and control access
- C9 management_control: compute management plane; topology requires site design
- C10 management_control: compute-fabric management plane
- C11 management_control: storage-fabric management plane
- C12 management_control: storage-system management plane
- C13 management_control: telemetry and monitoring
- C14 management_control: power and cooling constraint
- C15 logical_data_flow: customer service and data connectivity
- C16 logical_data_flow: external handoff - engineering input required

Not authoritative for detailed design, port maps, cable schedules, installation or procurement.

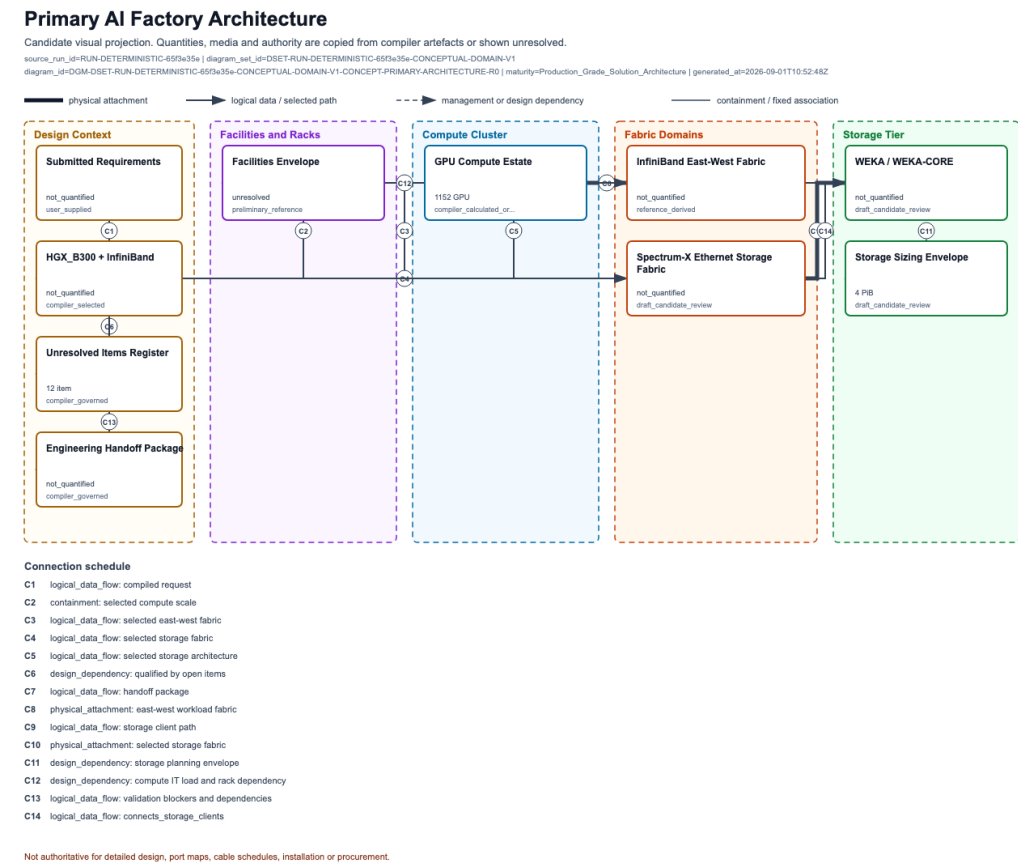
Conceptual Logical and Control Flows

Purpose: Conceptual Logical and Control Flows

How to read this figure: This Conceptual Logical and Control Flows presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.21 Figure 4: Primary AI Factory Architecture



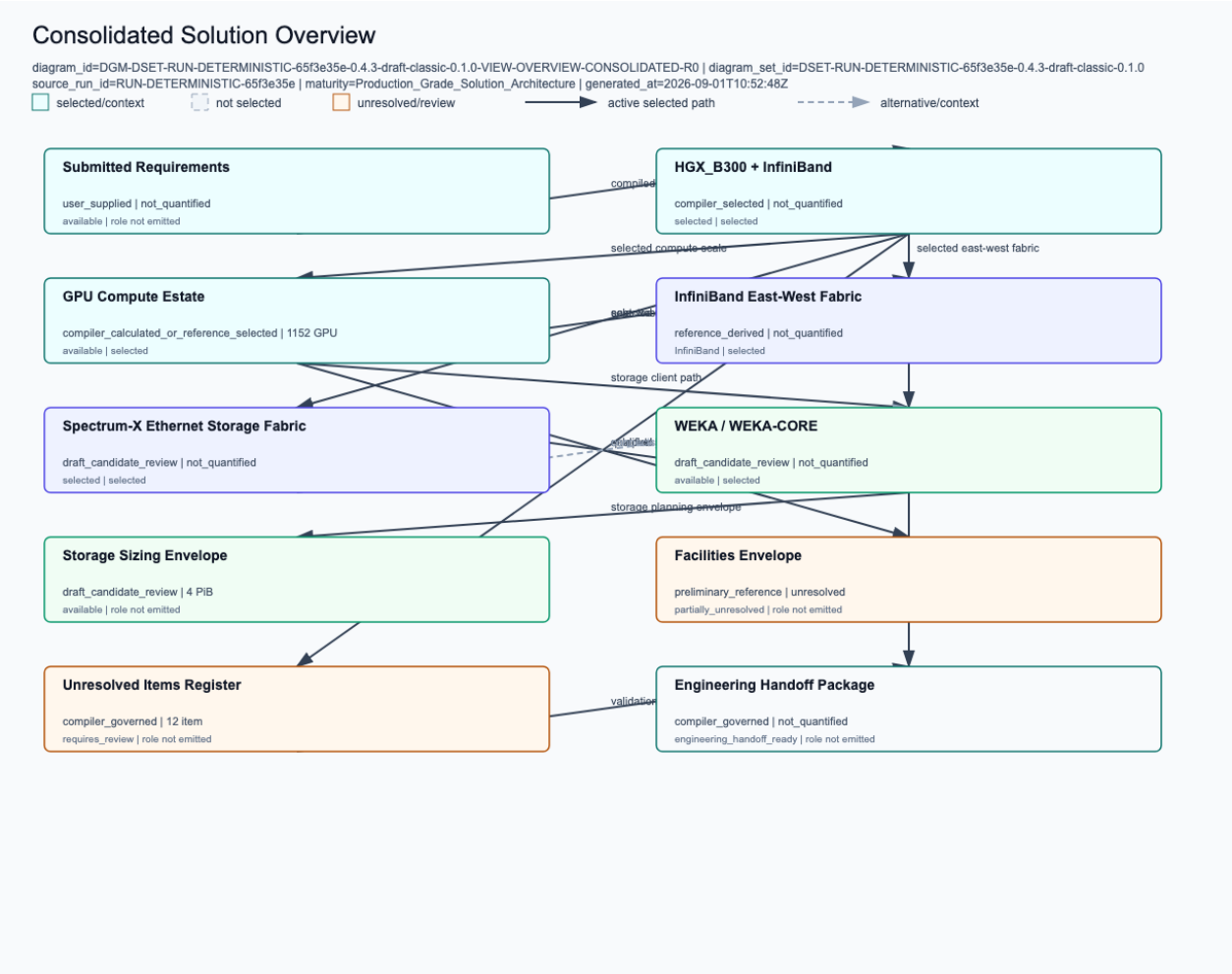
Primary AI Factory Architecture

Purpose: Primary AI Factory Architecture

How to read this figure: This Primary AI Factory Architecture presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.22 Figure 5: Consolidated Solution Overview



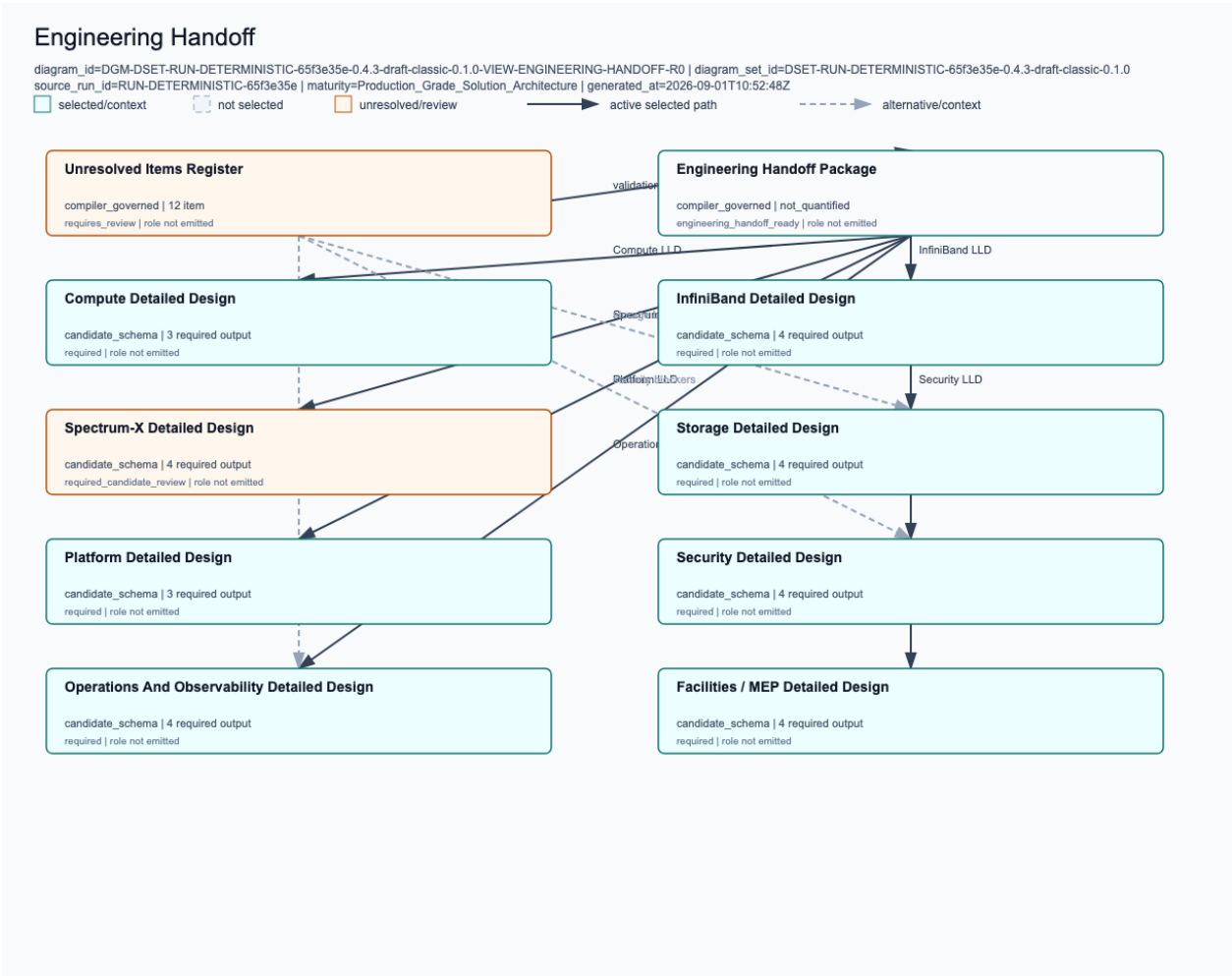
Consolidated Solution Overview

Purpose: Consolidated Solution Overview

How to read this figure: This Consolidated Solution Overview presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.23 Figure 6: Engineering Handoff



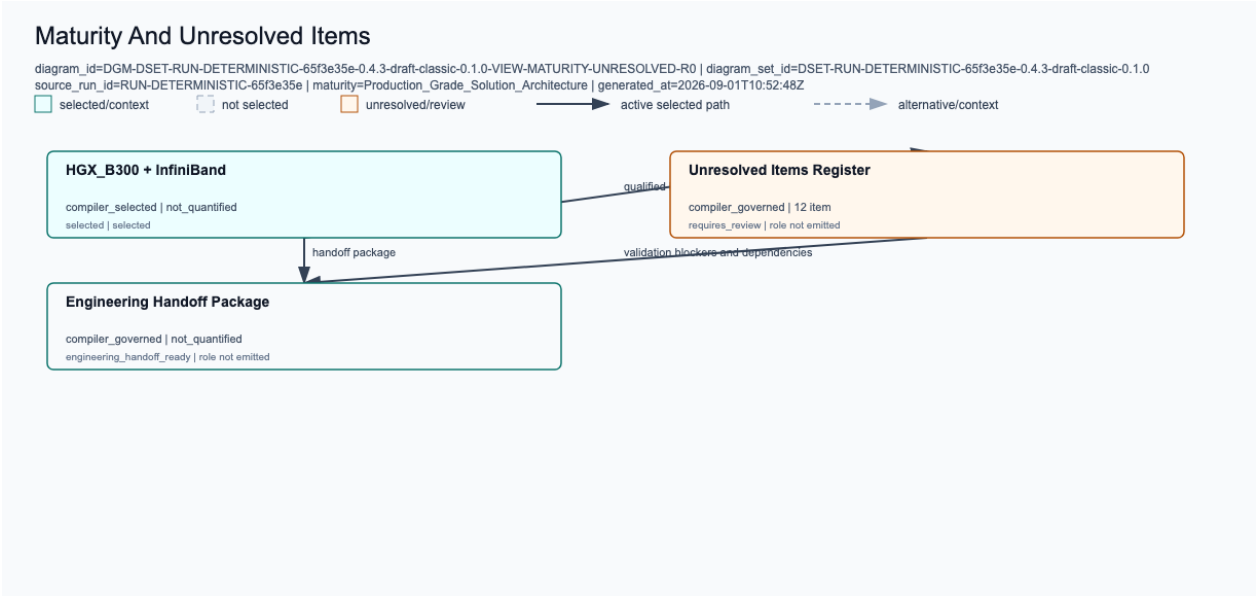
Engineering Handoff

Purpose: Engineering Handoff

How to read this figure: This Engineering Handoff presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.24 Figure 7: Maturity And Unresolved Items



Maturity And Unresolved Items

Purpose: Maturity And Unresolved Items

How to read this figure: This Maturity And Unresolved Items presents its governed architecture domains and explicitly labelled relationships.

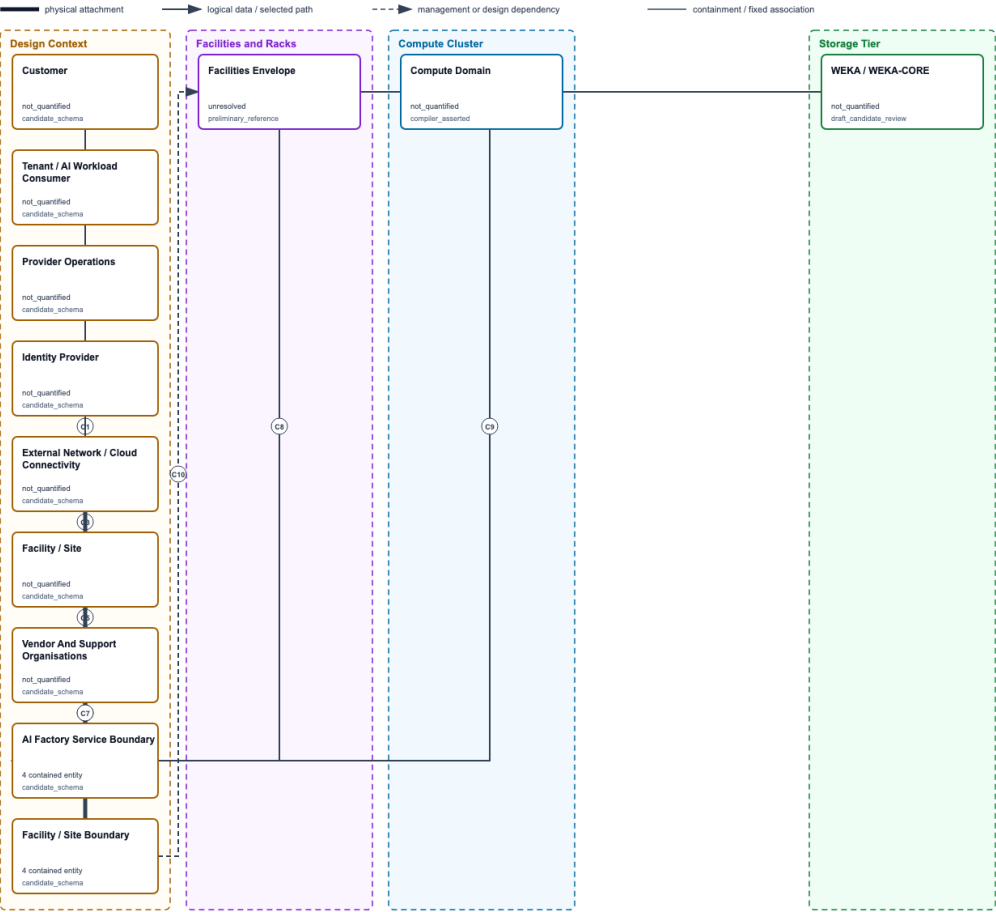
Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.25 Figure 8: Solution And Enterprise Context

Solution And Enterprise Context

Candidate visual projection. Quantities, media and authority are copied from compiler artefacts or shown unresolved.

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Connection schedule

- C1 logical_data_flow: AI Factory service
- C2 logical_data_flow: tenant service
- C3 management_control: provider operations
- C4 logical_data_flow: identity dependency
- C5 physical_attachment: ingress / egress
- C6 physical_attachment: site boundary
- C7 logical_data_flow: vendor support
- C8 containment: compute service
- C9 containment: storage service
- C10 design_dependency: facility envelope

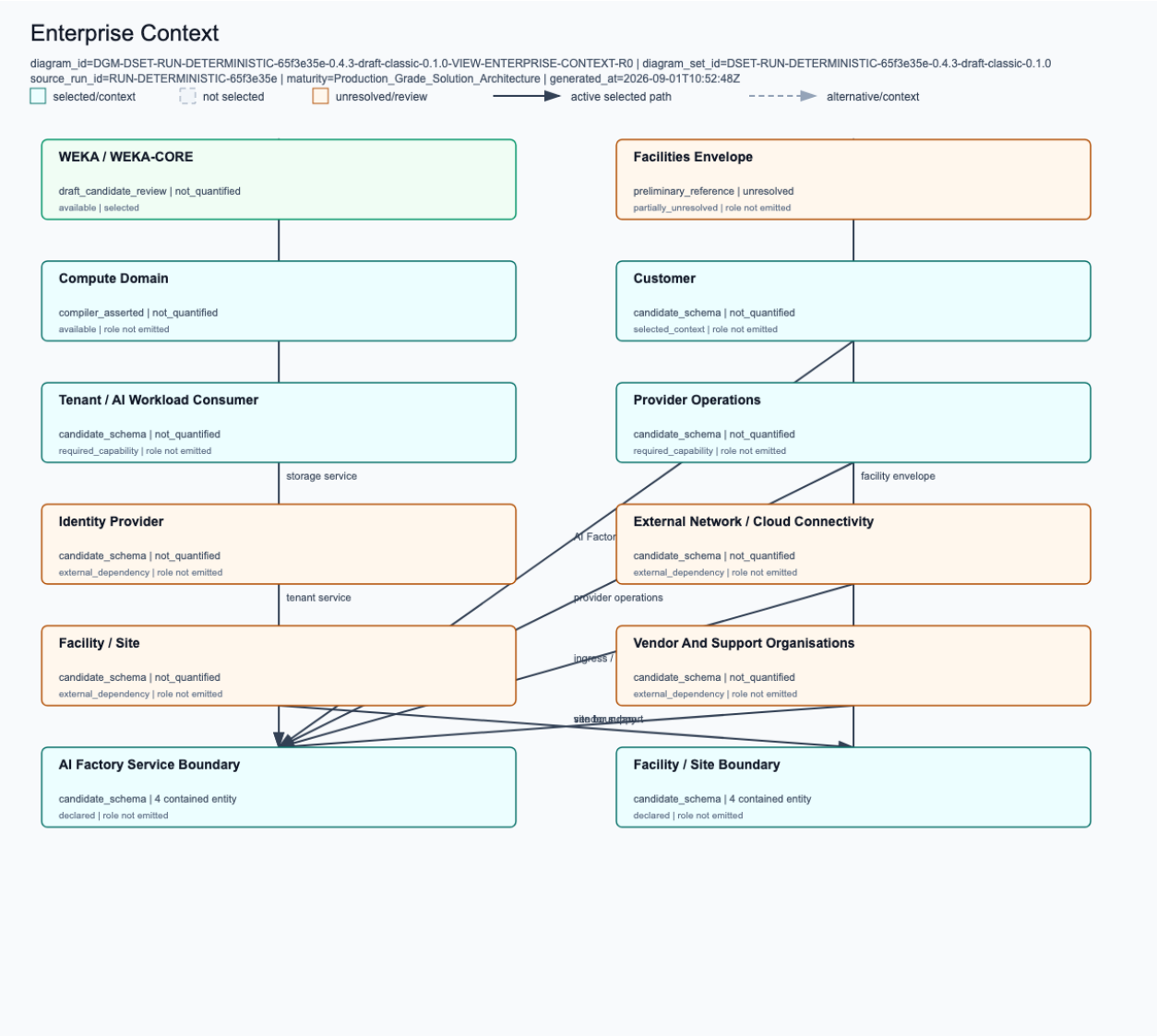
Not authoritative for detailed design, port maps, cable schedules, installation or procurement.

Purpose: Solution And Enterprise Context

How to read this figure: This Solution And Enterprise Context presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.26 Figure 9: Enterprise Context



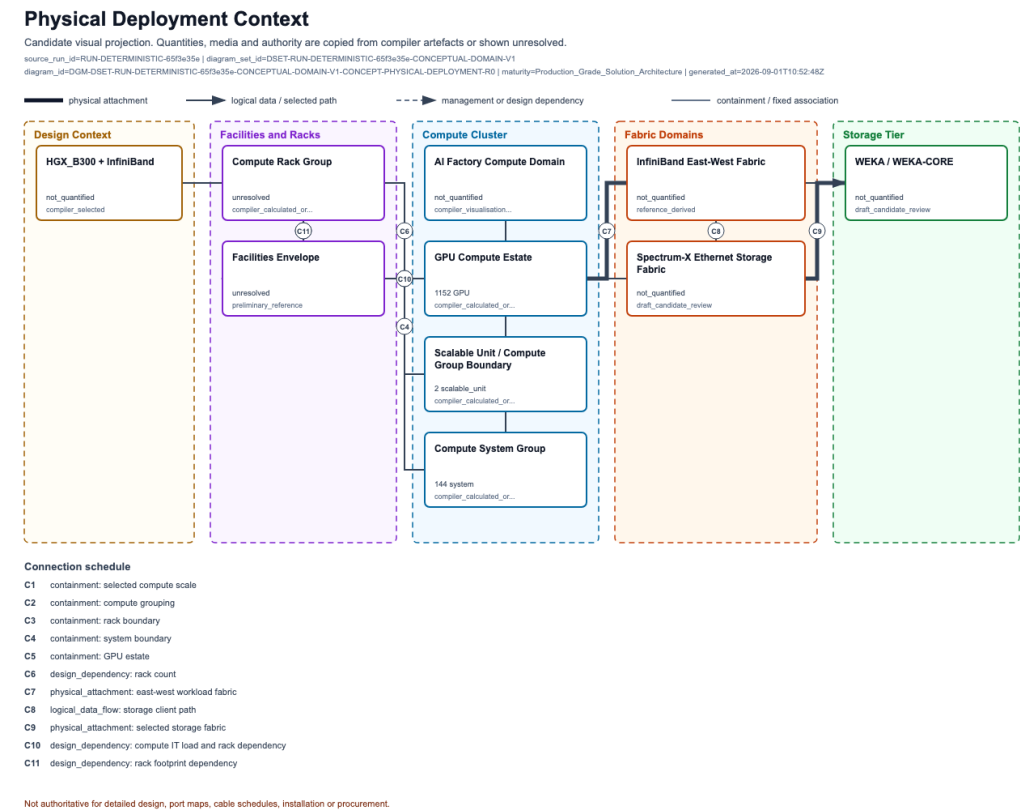
Enterprise Context

Purpose: Enterprise Context

How to read this figure: This Enterprise Context presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.27 Figure 10: Physical Deployment Context



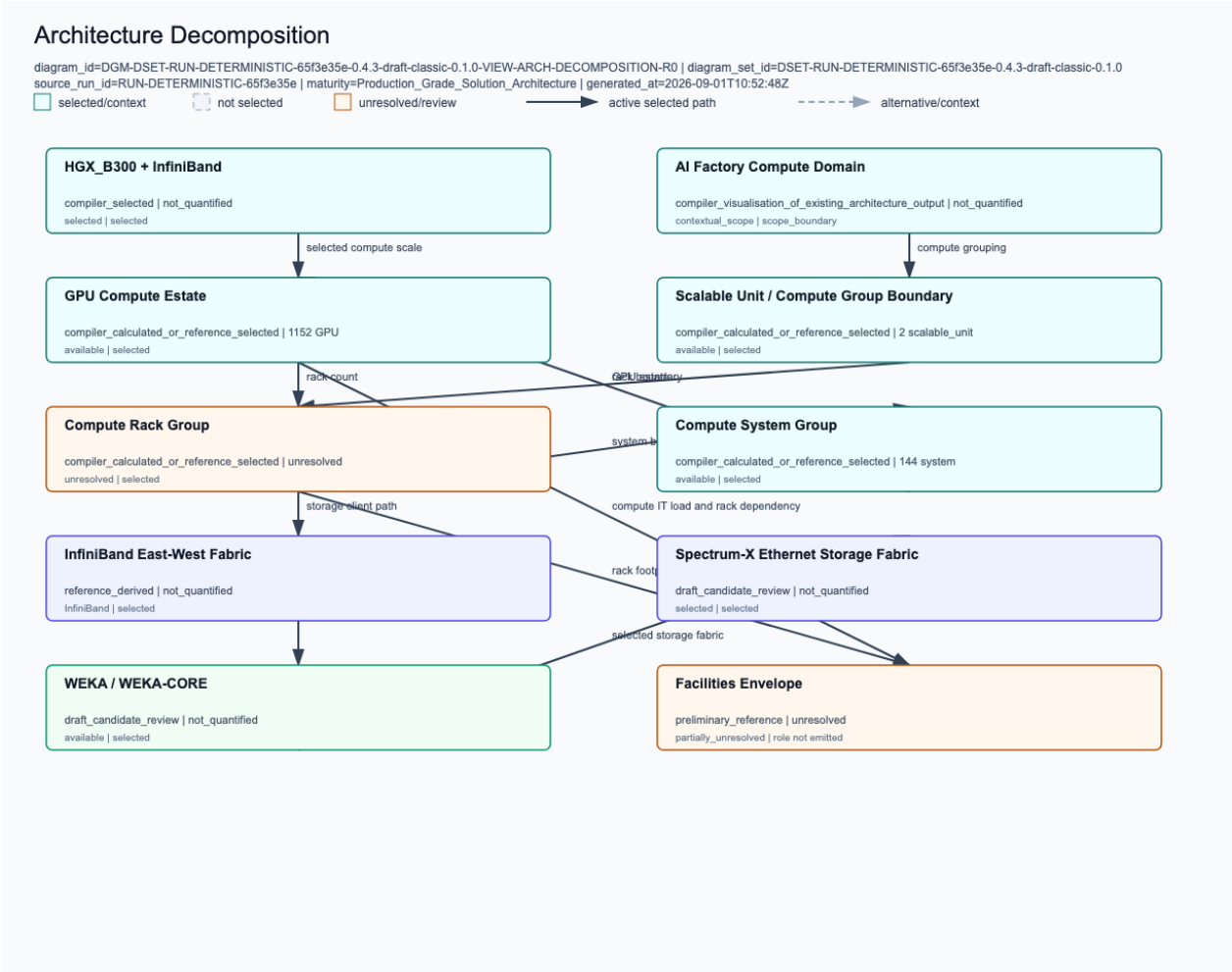
Physical Deployment Context

Purpose: Physical Deployment Context

How to read this figure: This Physical Deployment Context presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.28 Figure 11: Architecture Decomposition



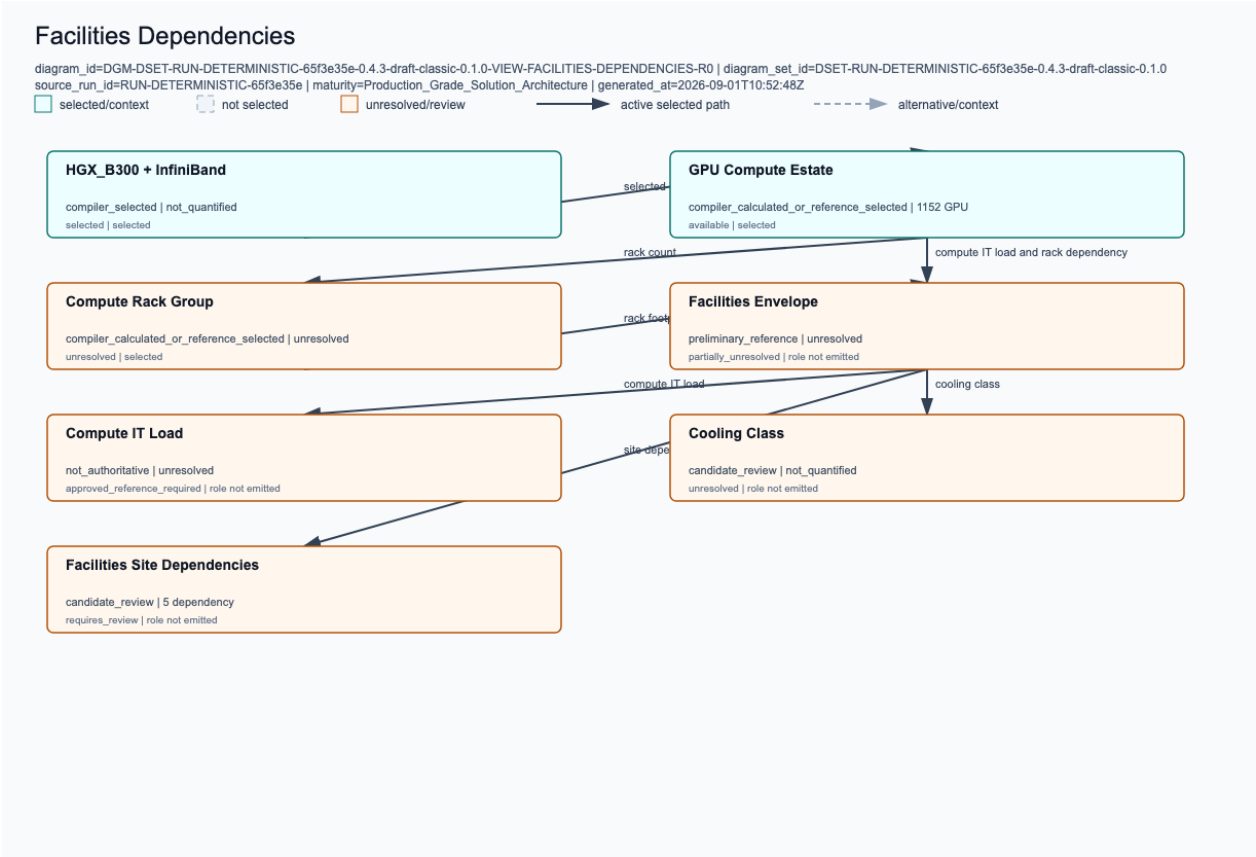
Architecture Decomposition

Purpose: Architecture Decomposition

How to read this figure: This Architecture Decomposition presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.29 Figure 12: Facilities Dependencies



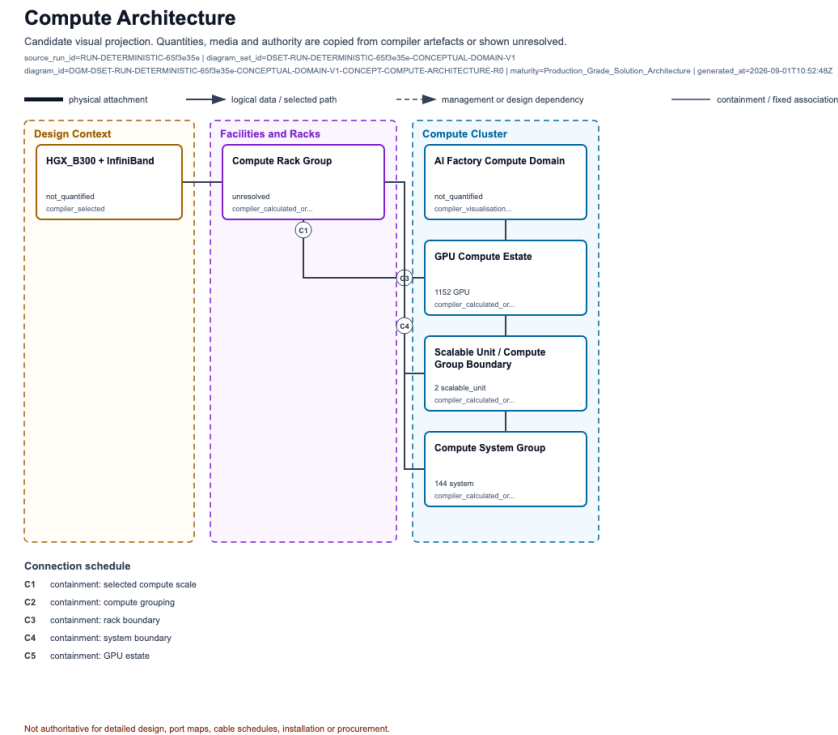
Facilities Dependencies

Purpose: Facilities Dependencies

How to read this figure: This Facilities Dependencies presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.30 Figure 13: Compute Architecture



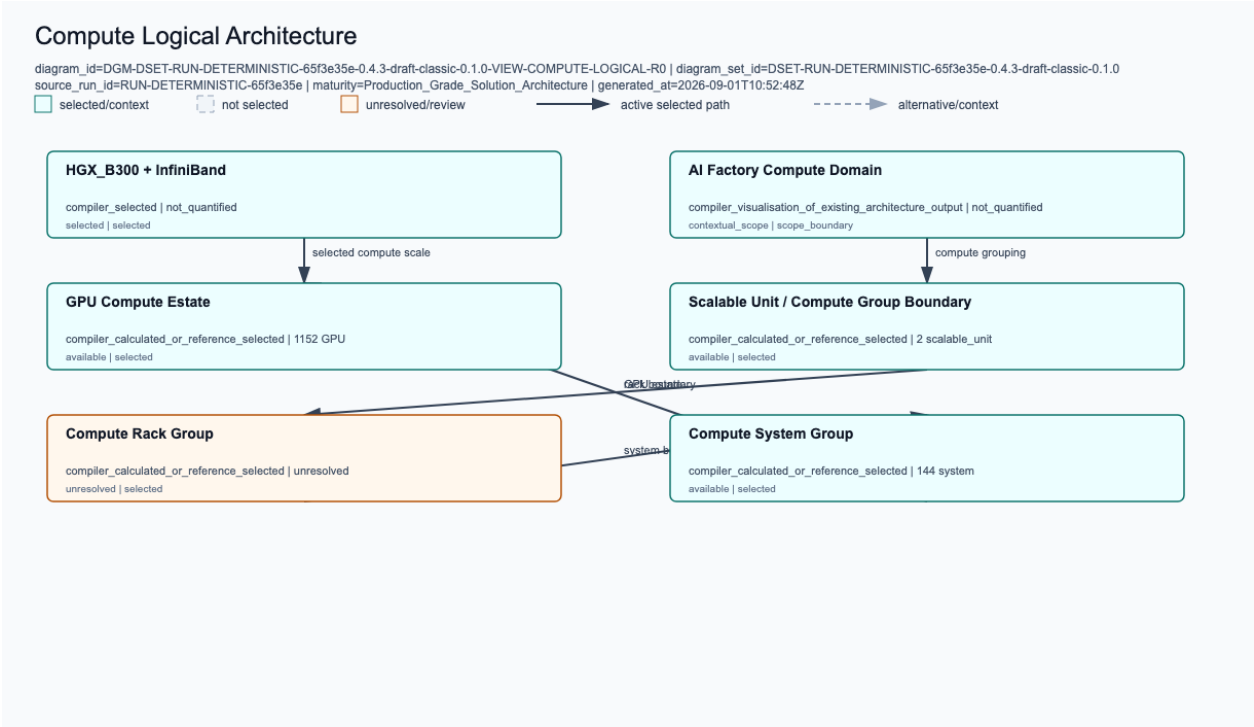
Compute Architecture

Purpose: Compute Architecture

How to read this figure: This Compute Architecture presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.31 Figure 14: Compute Logical Architecture



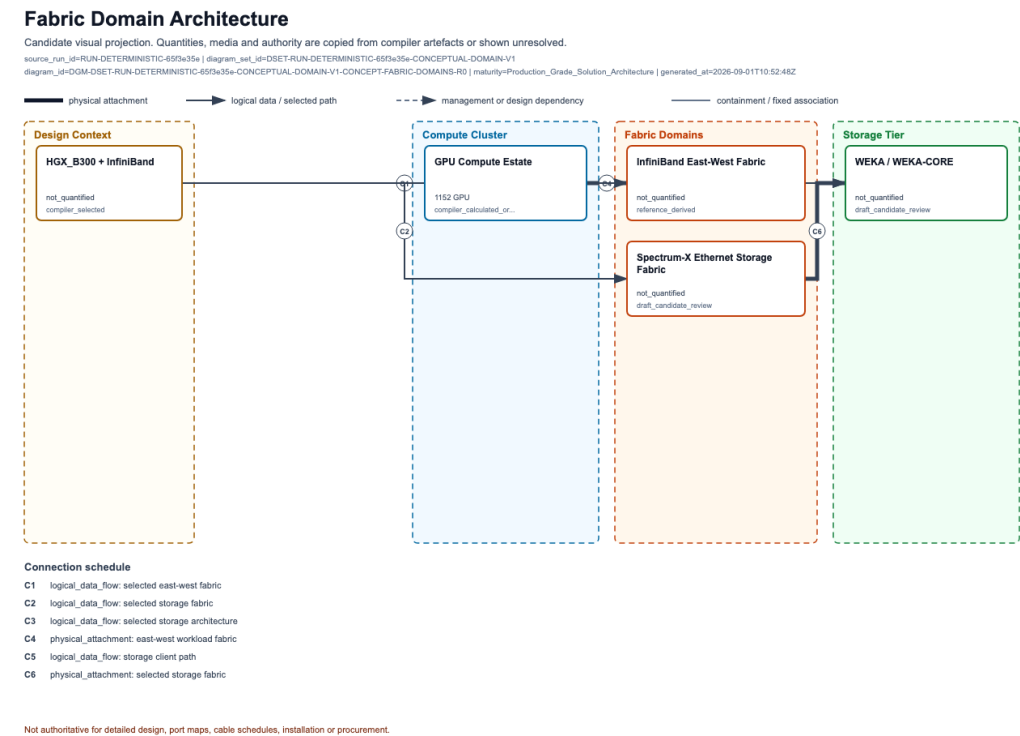
Compute Logical Architecture

Purpose: Compute Logical Architecture

How to read this figure: This Compute Logical Architecture presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.32 Figure 15: Fabric Domain Architecture



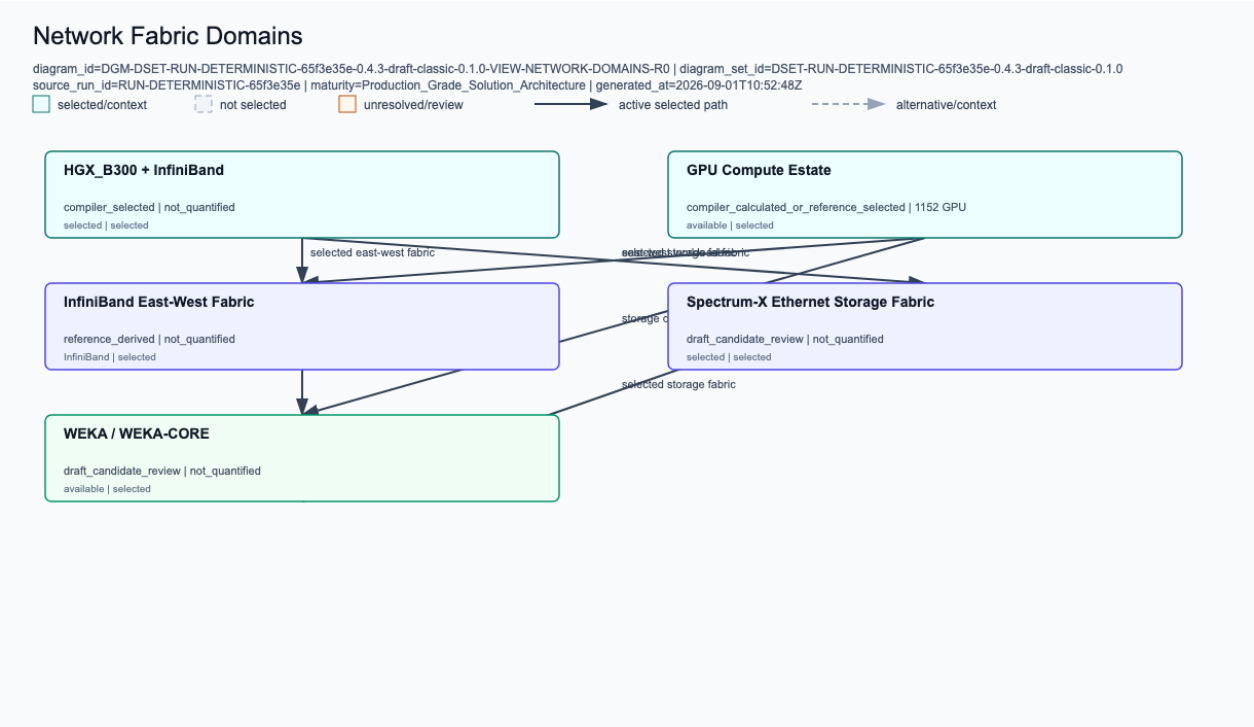
Fabric Domain Architecture

Purpose: Fabric Domain Architecture

How to read this figure: This Fabric Domain Architecture presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.33 Figure 16: Network Fabric Domains



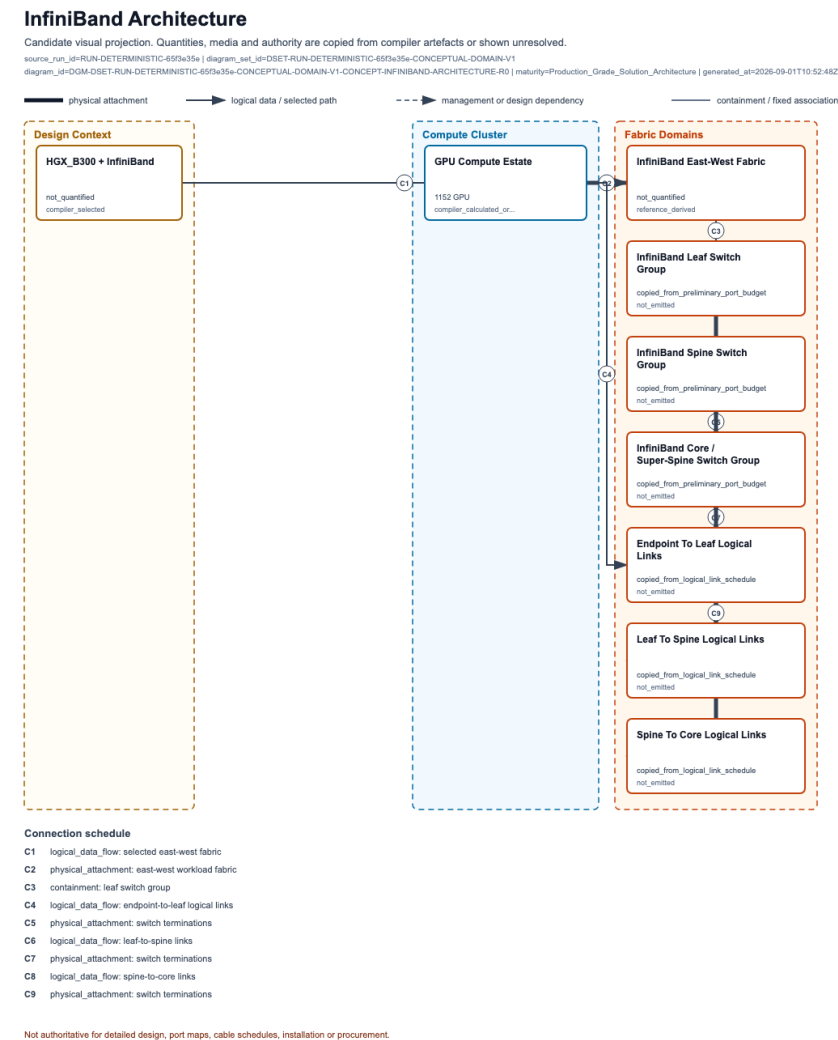
Network Fabric Domains

Purpose: Network Fabric Domains

How to read this figure: This Network Fabric Domains presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.34 Figure 17: InfiniBand Architecture



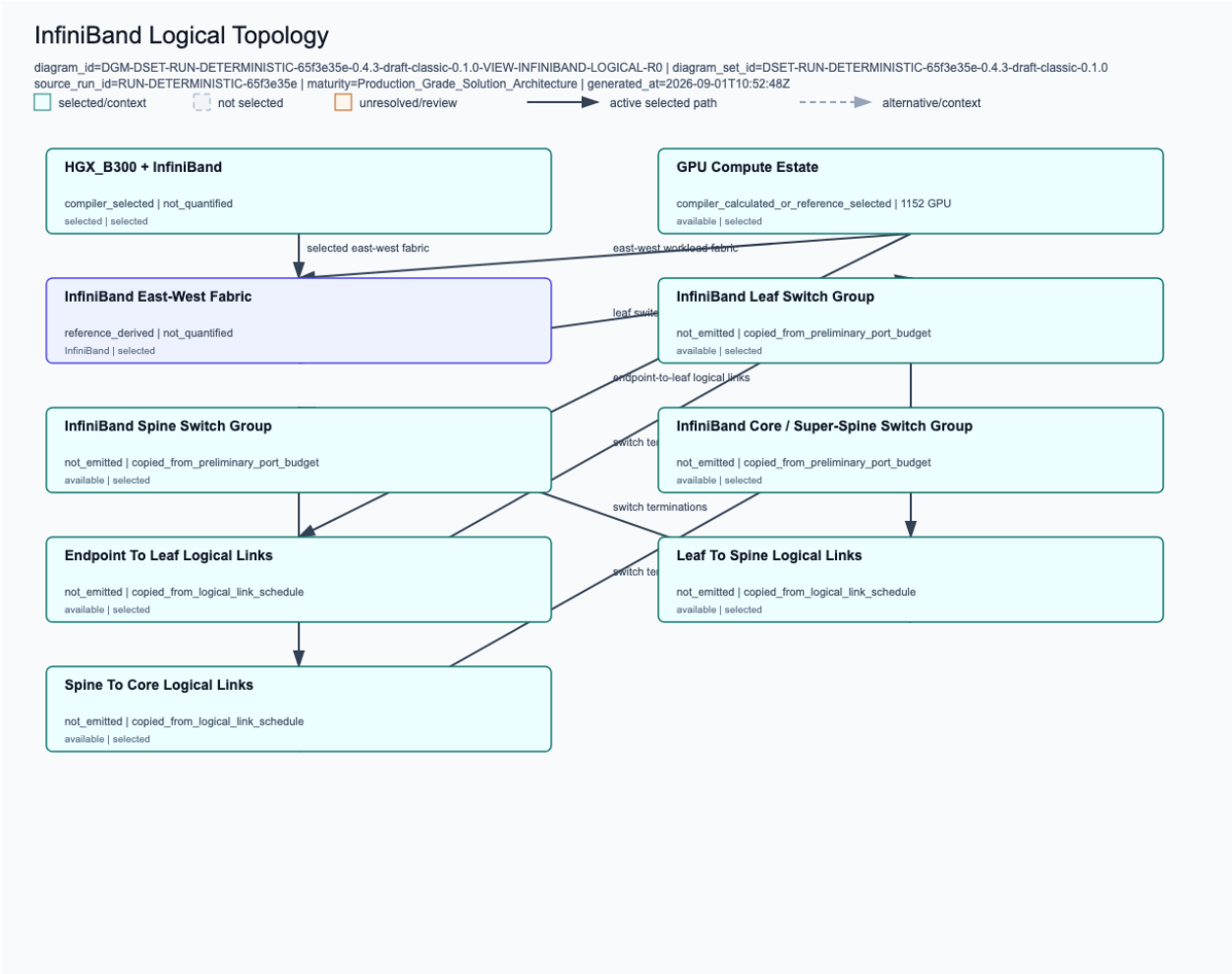
InfiniBand Architecture

Purpose: InfiniBand Architecture

How to read this figure: This InfiniBand Architecture presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.35 Figure 18: InfiniBand Logical Topology



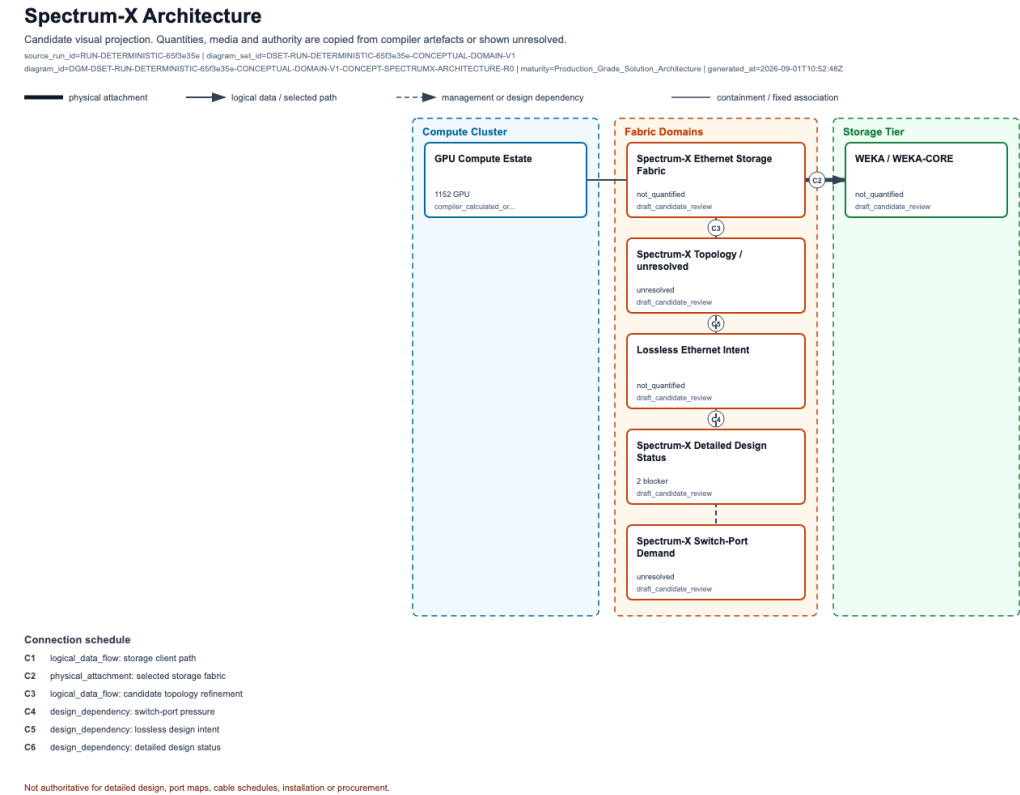
InfiniBand Logical Topology

Purpose: InfiniBand Logical Topology

How to read this figure: This InfiniBand Logical Topology presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.36 Figure 19: Spectrum-X Architecture



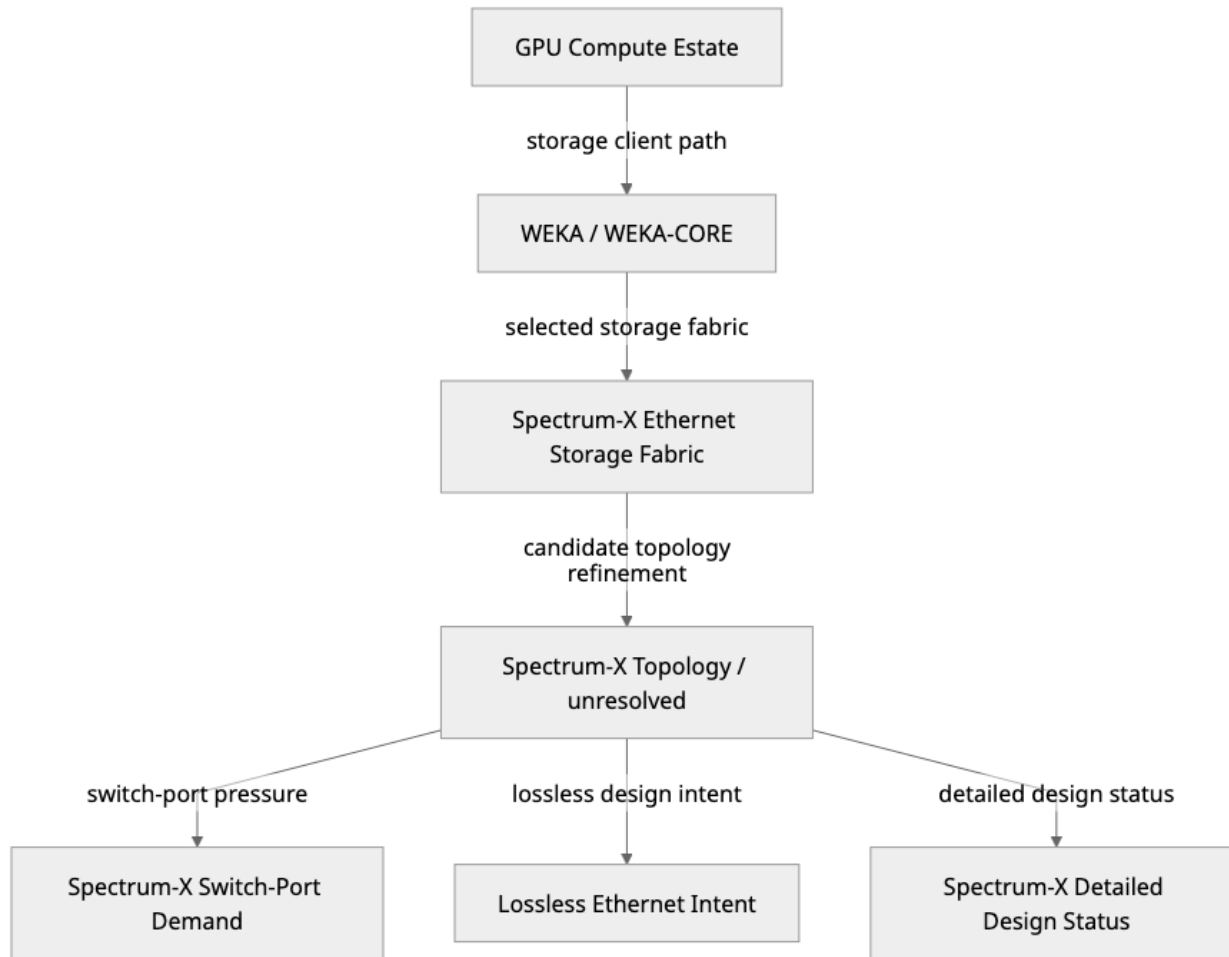
Spectrum-X Architecture

Purpose: Spectrum-X Architecture

How to read this figure: This Spectrum-X Architecture presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.37 Figure 20: Spectrum-X Logical Topology



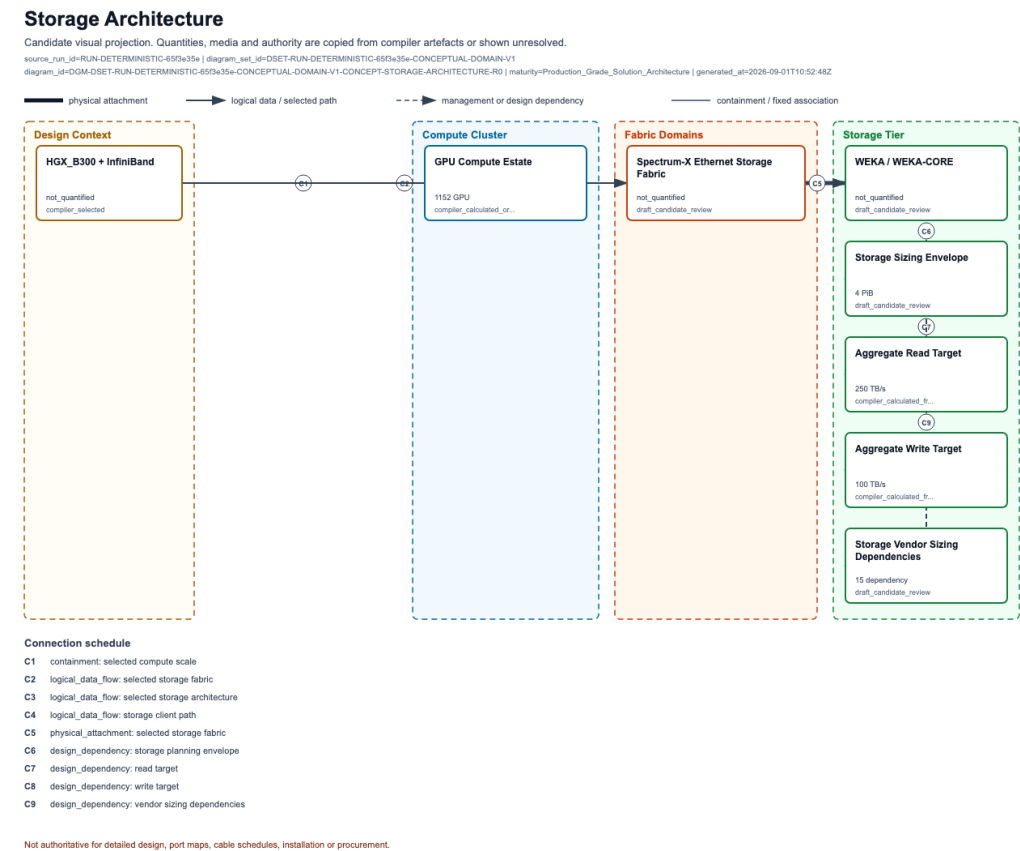
Spectrum-X Logical Topology

Purpose: Spectrum-X Logical Topology

How to read this figure: This Spectrum-X Logical Topology presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.38 Figure 21: Storage Architecture



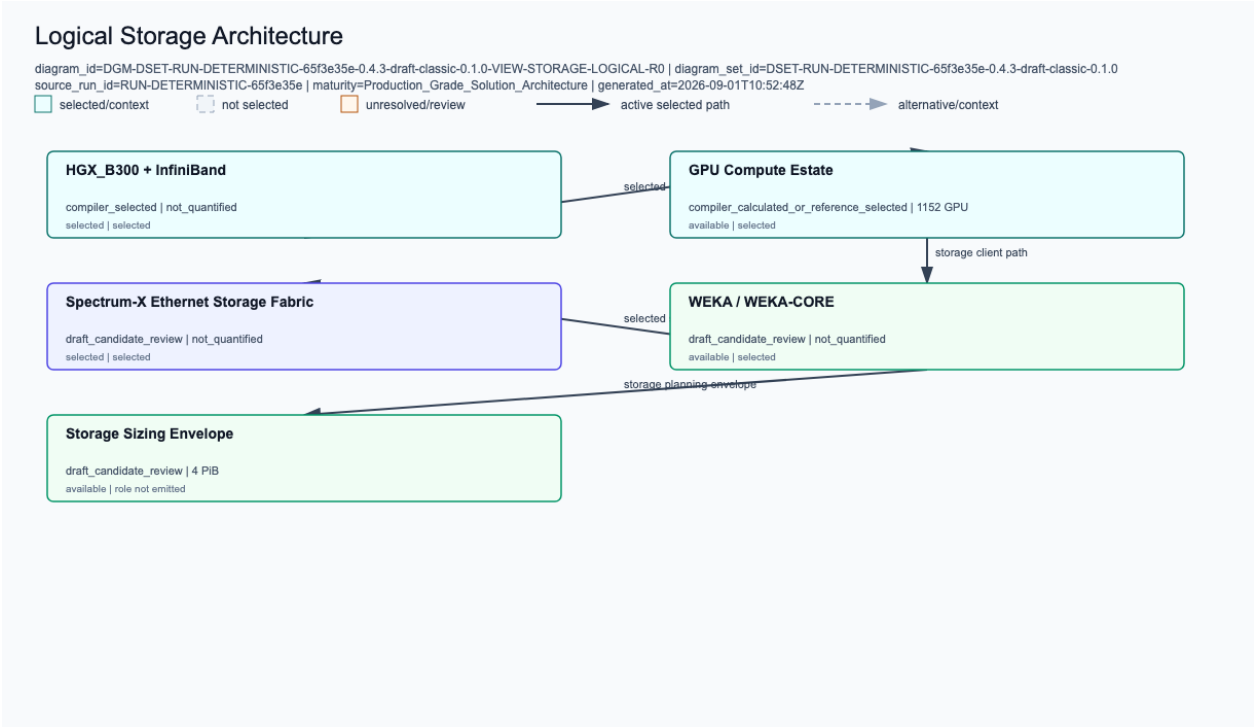
Storage Architecture

Purpose: Storage Architecture

How to read this figure: This Storage Architecture presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.39 Figure 22: Logical Storage Architecture



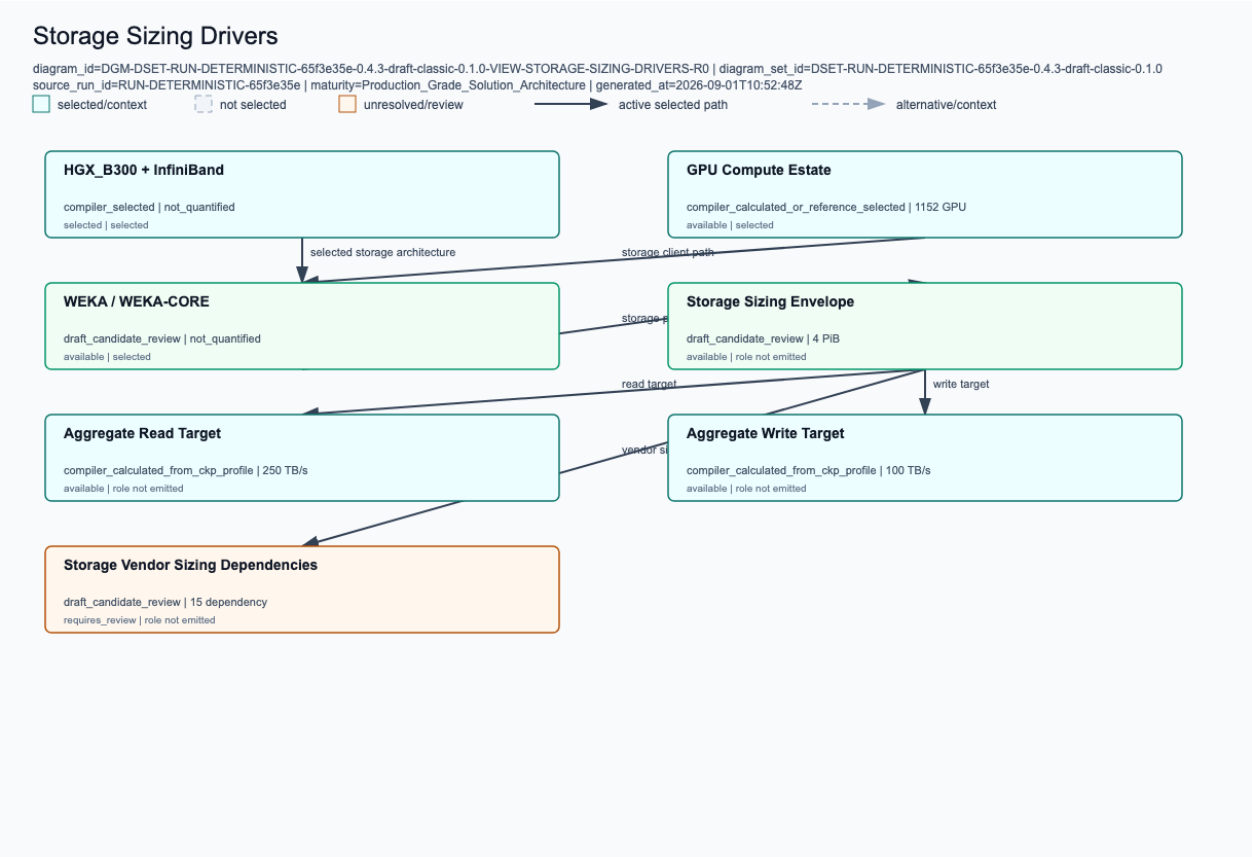
Logical Storage Architecture

Purpose: Logical Storage Architecture

How to read this figure: This Logical Storage Architecture presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.40 Figure 23: Storage Sizing Drivers



Storage Sizing Drivers

Purpose: Storage Sizing Drivers

How to read this figure: This Storage Sizing Drivers presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

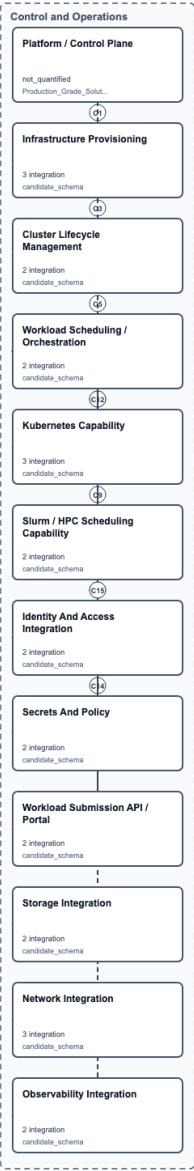
2.41 Figure 24: Platform And Control Plane

Platform And Control Plane

Candidate visual projection. Quantities, media and authority are copied from compiler artefacts or shown unresolved.

source_run_is=RUN-DETERMINISTIC-65f3a35e | diagram_wt_is=DSET-RUN-DETERMINISTIC-65f3a35e-CONCEPTUAL-DOMAIN-V1
diagram_is=DGM-DSET-RUN-DETERMINISTIC-65f3a35e-CONCEPTUAL-DOMAIN-V1-CONCEPT-PLATFORM-CONTROL-R0 | maturity=Production_Grade_Solution_Architecture | generated_at=2026-09-01T10:52:48Z

physical attachment logical data / selected path management or design dependency containment / fixed association



Connection schedule

- C1 design_dependency: provisioning
- C2 design_dependency: lifecycle
- C3 design_dependency: scheduler
- C4 design_dependency: Kubernetes capability
- C5 design_dependency: Slurm capability
- C6 design_dependency: identity
- C7 design_dependency: secrets / policy
- C8 design_dependency: workload submission
- C9 design_dependency: storage integration

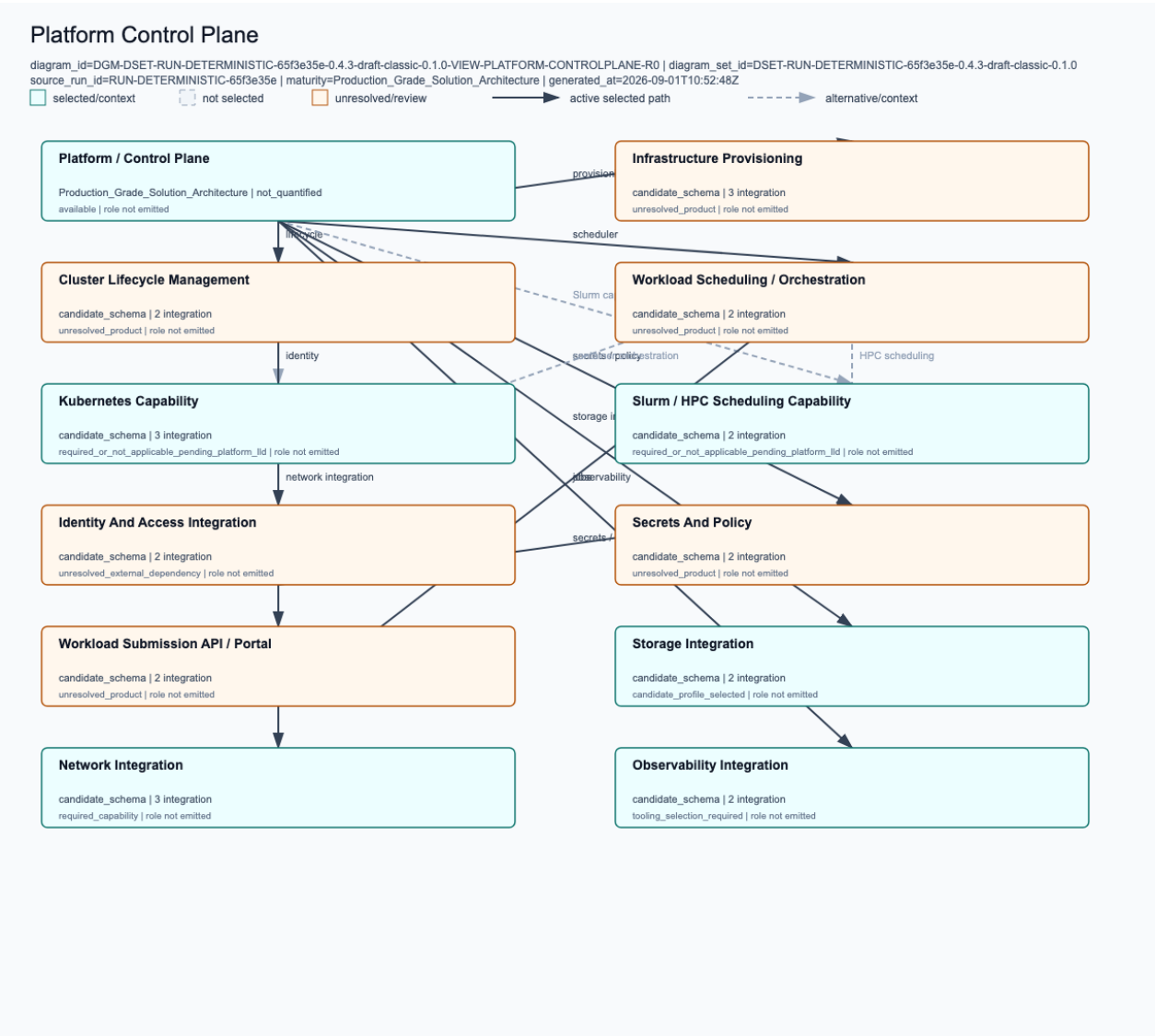
Platform And Control Plane

Purpose: Platform And Control Plane

How to read this figure: This Platform And Control Plane presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.42 Figure 25: Platform Control Plane



Platform Control Plane

Purpose: Platform Control Plane

How to read this figure: This Platform Control Plane presents its governed architecture domains and explicitly labelled relationships.

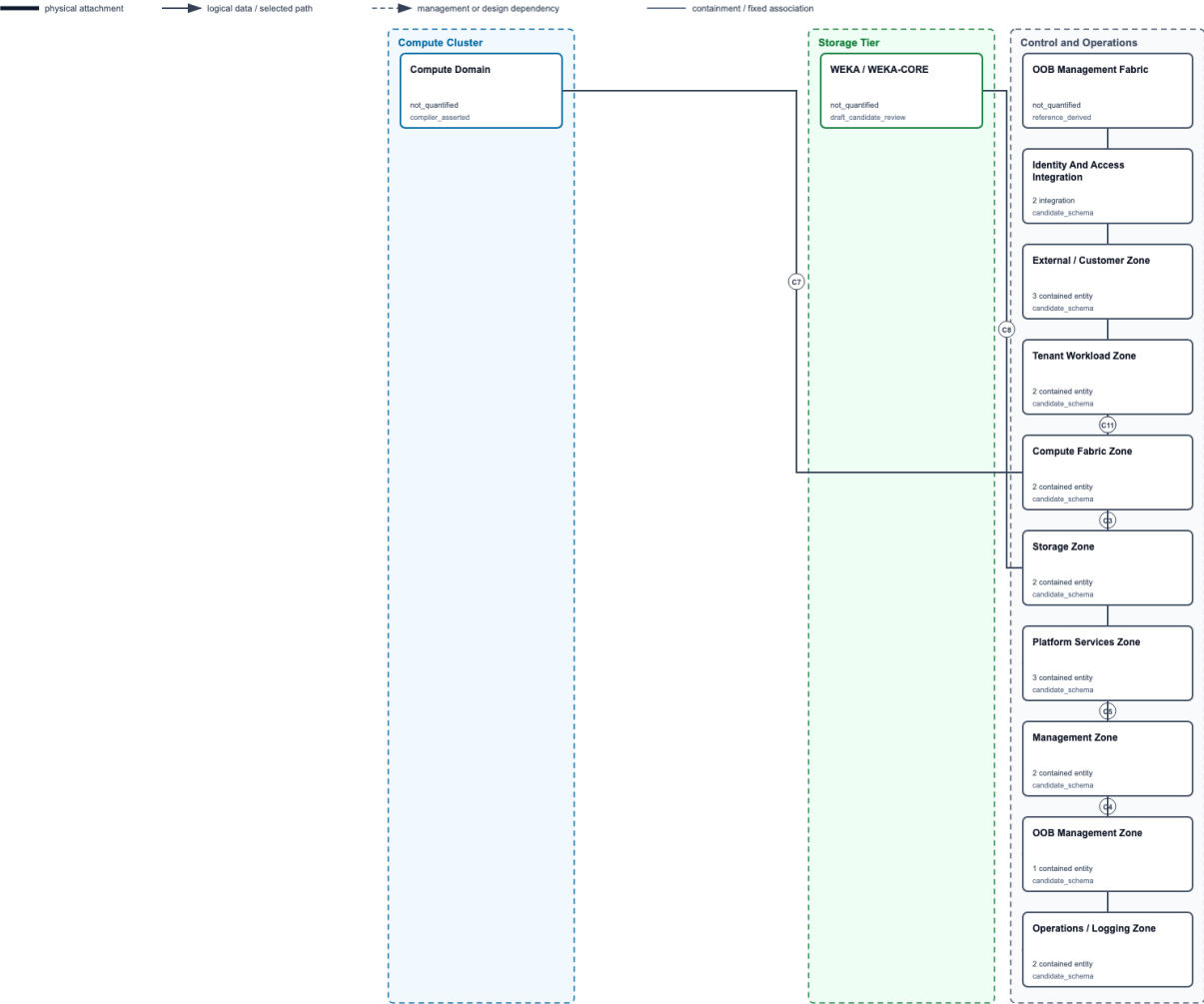
Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.43 Figure 26: Security And Trust Boundaries

Security And Trust Boundaries

Candidate visual projection. Quantities, media and authority are copied from compiler artefacts or shown unresolved.

source_run_is=RUN-DETERMINISTIC-6503a35e | diagram_wt_is=DSET-RUN-DETERMINISTIC-6503a35e-CONCEPTUAL-DOMAIN-V1
diagram_is=DGM-DSET-RUN-DETERMINISTIC-6503a35e-CONCEPT-DOMAIN-V1-CONCEPT-SECURITY-TRUST-R0 | maturity=Production_Grade_Solution_Architecture | generated_at=2026-09-01T10:52:48Z



Connection schedule

- C1 logical_data_flow: workload_submission
- C2 logical_data_flow: control_plane
- C3 logical_data_flow: storage_data_path
- C4 logical_data_flow: privileged_administration
- C5 logical_data_flow: telemetry
- C6 logical_data_flow: telemetry
- C7 containment: compute
- C8 containment: storage
- C9 containment: identity
- C10 containment: OOB
- C11 logical_data_flow: workload traffic

Security And Trust Boundaries

Purpose: Security And Trust Boundaries

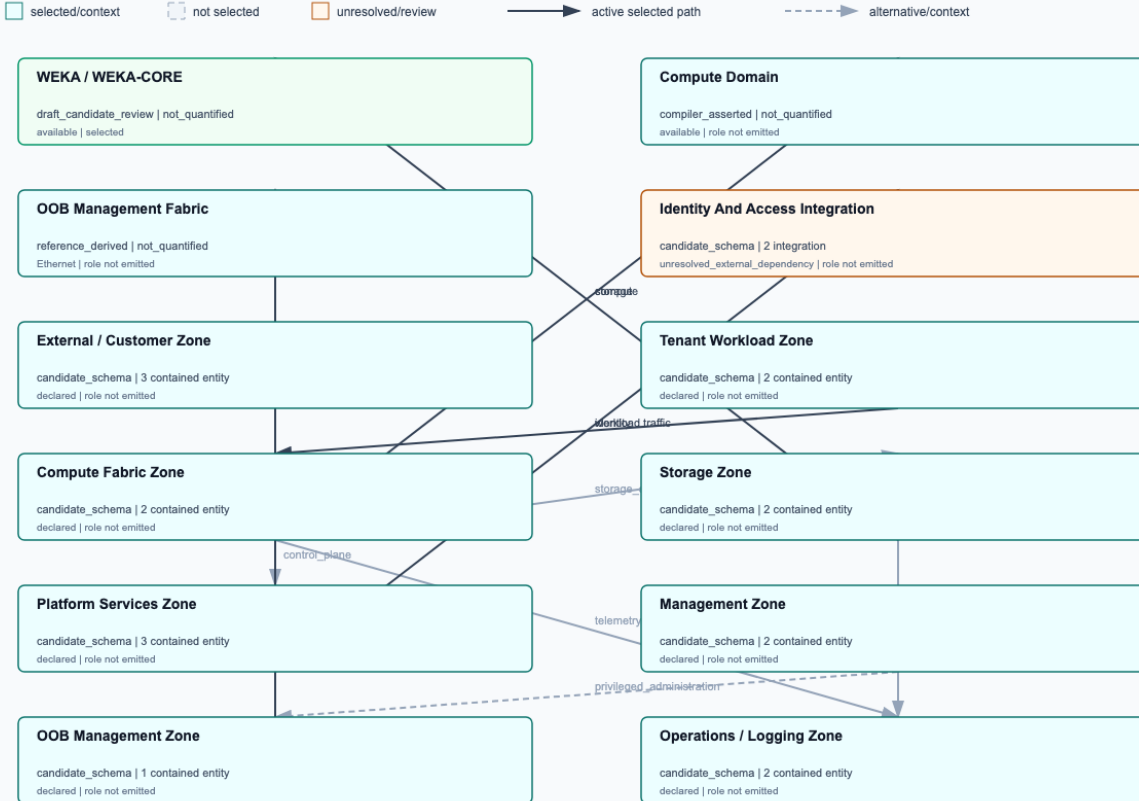
How to read this figure: This Security And Trust Boundaries presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.44 Figure 27: Security Zones



diagram_id=DGM-DSET-RUN-DETERMINISTIC-65f3e35e-0.4.3-draft-classic-0.1.0-VIEW-SECURITY-ZONES-R0 | diagram_set_id=DSET-RUN-DETERMINISTIC-65f3e35e-0.4.3-draft-classic-0.1.0
source_run_id=RUN-DETERMINISTIC-65f3e35e | maturity=Production_Grade_Solution_Architecture | generated_at=2026-09-01T10:52:48Z



Security Zones

Purpose: Security Zones

How to read this figure: This Security Zones presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

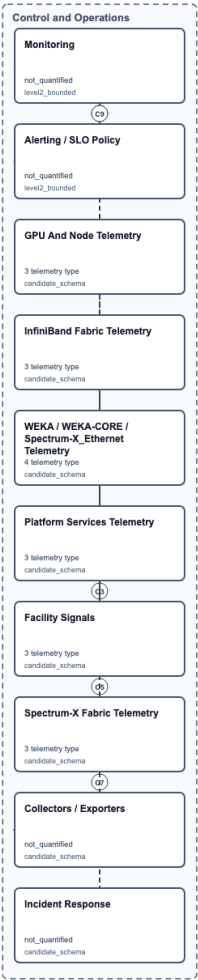
2.45 Figure 28: Operations And Observability

Operations And Observability

Candidate visual projection. Quantities, media and authority are copied from compiler artefacts or shown unresolved.

source_run_is=RUN-DETERMINISTIC-65f3a35e | diagram_wt_is=DSET-RUN-DETERMINISTIC-65f3a35e-CONCEPTUAL-DOMAIN-V1
diagram_is=DGM-DSET-RUN-DETERMINISTIC-65f3a35e-CONCEPTUAL-DOMAIN-V1-CONCEPT-OPERATIONS-R0 | maturity=Production_Grade_Solution_Architecture | generated_at=2026-09-01T10:52:48Z

physical attachment logical data / selected path management or design dependency containment / fixed association



Connection schedule

- C1 design_dependency: SLO alerting
- C2 management_control: telemetry
- C3 management_control: telemetry
- C4 management_control: telemetry
- C5 management_control: telemetry
- C6 management_control: telemetry
- C7 management_control: telemetry
- C8 management_control: metrics_logs_events
- C9 management_control: alerting
- C10 management_control: incident_escalation

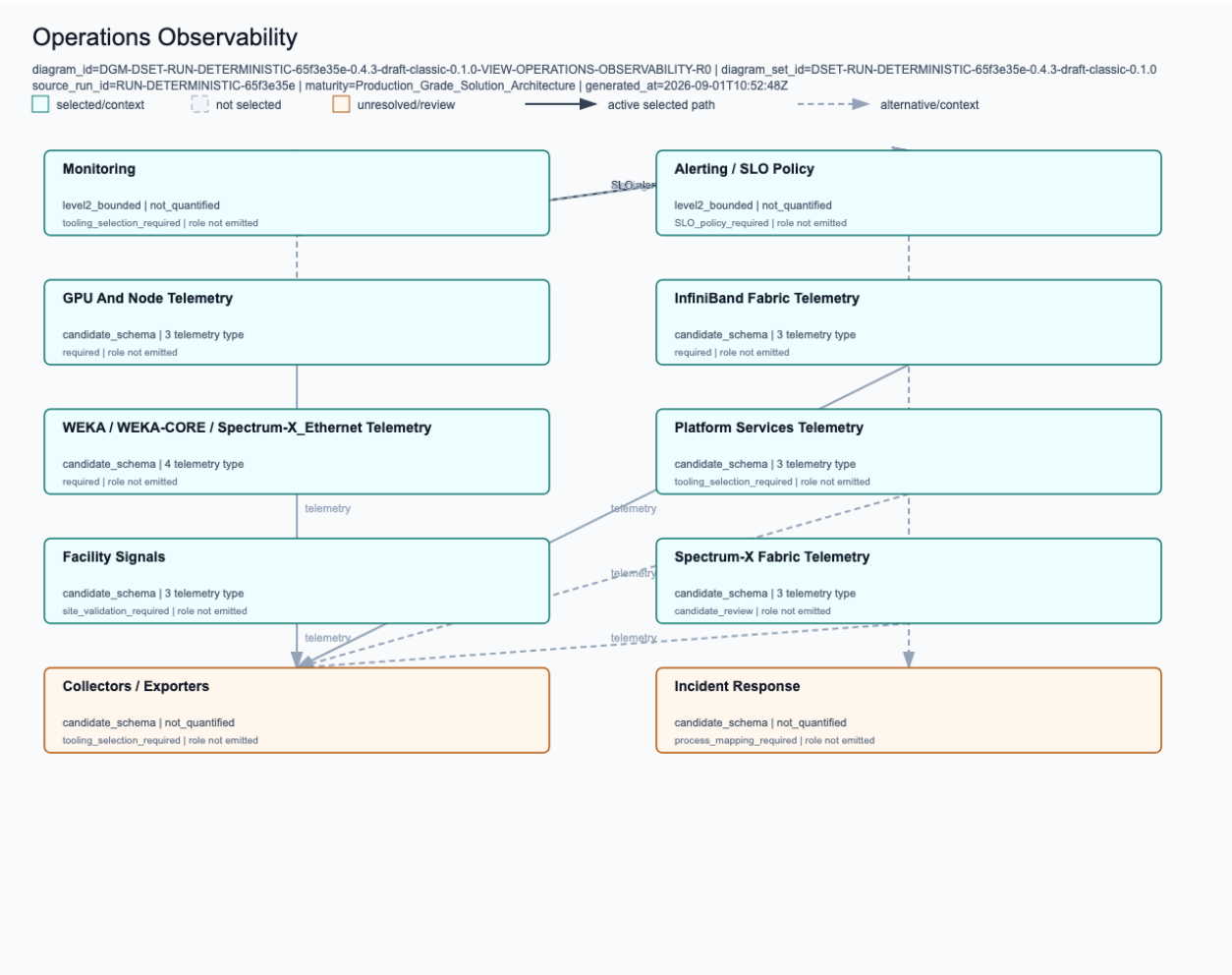
Not authoritative for detailed design, port maps, cable schedules, installation or procurement.

Purpose: Operations And Observability

How to read this figure: This Operations And Observability presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.46 Figure 29: Operations Observability

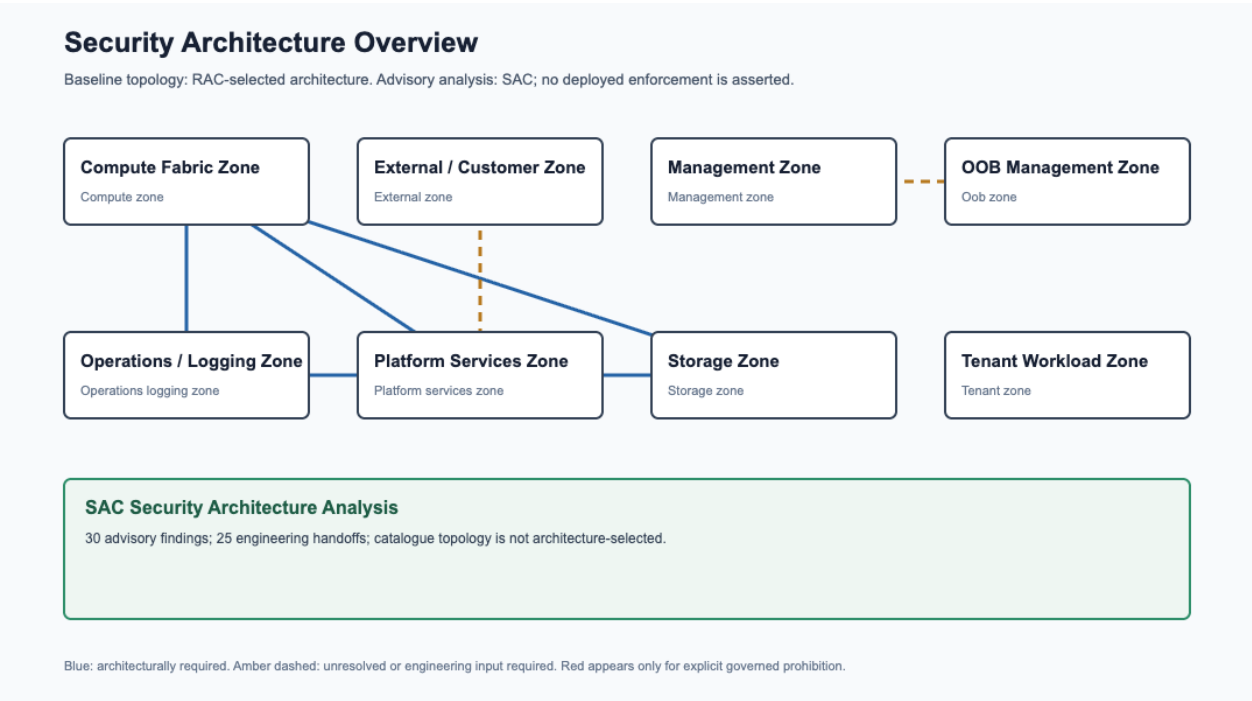


Purpose: Operations Observability

How to read this figure: This Operations Observability presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.47 Figure 30: Security Architecture Overview



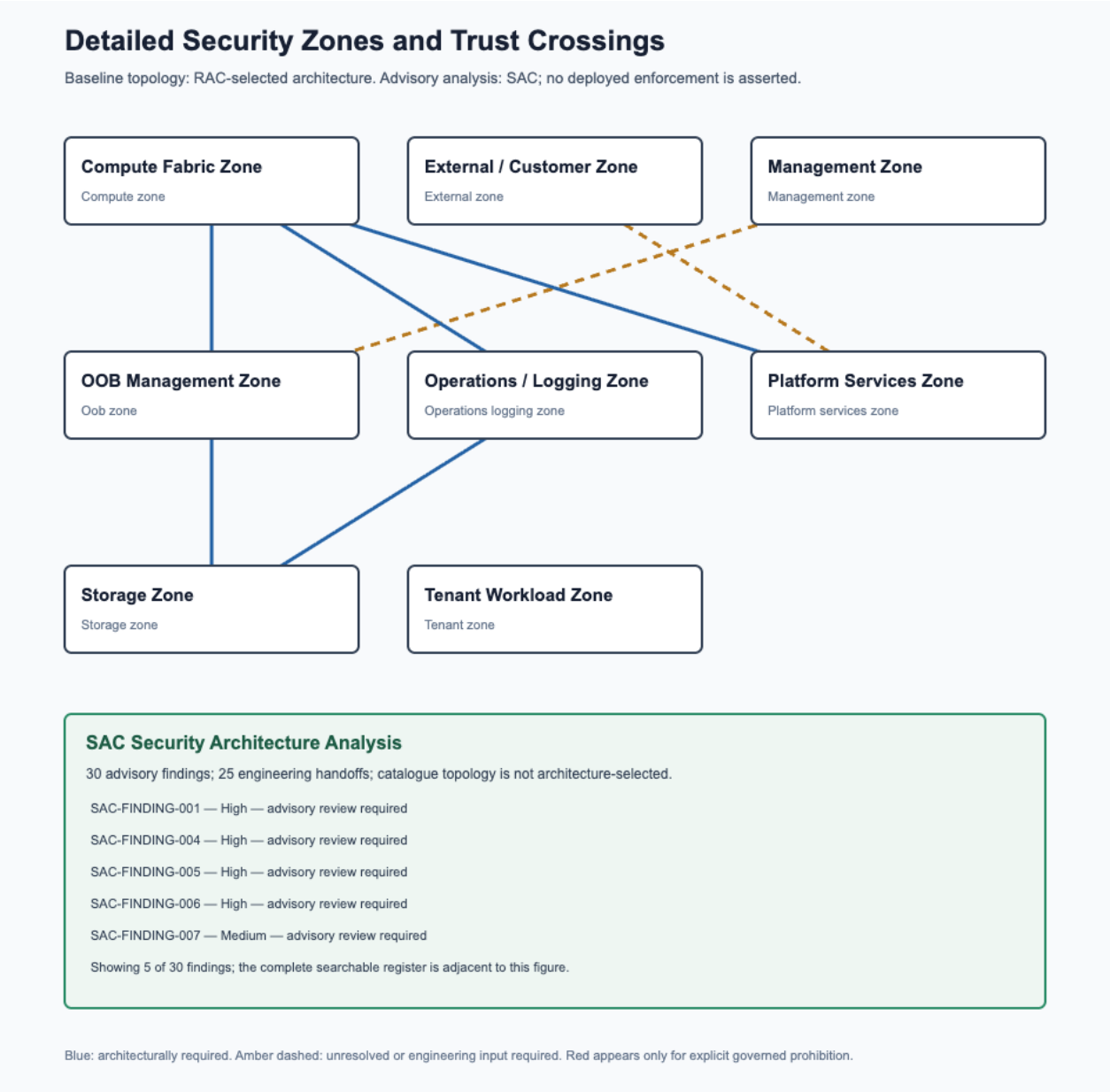
Security Architecture Overview

Purpose: Security Architecture Overview

How to read this figure: This Security Architecture Overview presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.48 Figure 31: Security Zones and Governed Crossings



Purpose: Security Zones and Governed Crossings

How to read this figure: This Security Zones and Governed Crossings presents its governed architecture domains and explicitly labelled relationships.

Limitations: Not procurement-ready, installation-ready, or a detailed design drawing.

2.49 Applicable Requirements Register

Requirement ID	Domain	Requirement	Applicability	Disposition
AF-BOM-SKEL-001	Bill of materials	BOM skeleton and preliminary BOM records shall expose quantity classification, evidence class and procurement readiness.	Applicable	Partially supported
AF-COMP-GENERIC-HGX-001	Compute	A generic HGX path shall not silently inherit DGX, GB300 or OEM-specific governed facts.	Applicable	Unresolved — engineering input required
AF-COMP-PLATFORM-001	Compute	The architecture request shall resolve to a supported compute platform or fail safely.	Applicable	Unresolved — engineering input required
AF-COMP-SCALE-001	Compute	The architecture request shall include an explicit target GPU count.	Applicable	Unresolved — engineering input required
AF-COMP-SCALE-ROUNDING-001	Compute	When target GPU count maps to a supported scale point by rounding or explicit candidate policy, the interpretation shall be emitted.	Applicable	Partially supported
AF-DD-BLOCKER-001	Architecture governance	Outputs shall list blockers before detailed design separately from blockers before procurement.	Applicable	Unresolved — engineering input required
AF-EXT-RESP-001	Architecture governance	Requirements satisfied by customer, operator, vendor, integrator or facility owner shall identify the external responsibility	Applicable	Partially supported

Requirement ID	Domain	Requirement	Applicability	Disposition
AF-FAC-SITE-001	Facilities	owner. Facility dependencies and unresolved site-specific inputs shall identify the facilities owner or validation discipline.	Applicable	Partially supported
AF-GOV-NOLLM-001	Architecture governance	Runtime LLM output shall not determine requirement applicability, support status or architecture selection.	Applicable	Unresolved — engineering input required
AF-GOV-NONPROC-001	Architecture governance	Planning BOM and architecture outputs shall distinguish non-procurement-ready records from procurement-ready line items.	Applicable	Partially supported
AF-GOV-QSAFETY-001	Architecture governance	Every quantitative field designated as governed shall identify its value state, calculation or source basis, evidence class, confidence and prohibited uses.	Applicable	Partially supported
AF-GOV-STATUS-001	Architecture governance	The generated architecture shall expose its maturity and authority status.	Applicable	Unresolved — engineering input required
AF-GOV-TRACE-001	Architecture governance	Requirement support claims shall trace to stable architecture capabilities, features, decisions, elements,	Applicable	Partially supported

Requirement ID	Domain	Requirement	Applicability	Disposition
		calculations or responsibilities rather than presentation labels.		
AF-GOV-UNKNOWN-001	Architecture governance	Unknown engineering quantities shall be emitted as null or unresolved records, not as fabricated zero values.	Applicable	Partially supported
AF-NET-EW-001	Network	The architecture shall declare the selected east-west compute fabric.	Applicable	Unresolved — engineering input required
AF-NET-IB-001	Network	An InfiniBand architecture path shall expose whether logical topology information is reference-use-only and detailed-design validation required.	Applicable	Unresolved — engineering input required
AF-NET-OOB-001	Out-of-band management	The architecture shall include an out-of-band or logically independent management path distinct from primary workload data paths.	Applicable	Unresolved — engineering input required
AF-NET-PORT-RECON-001	Network	Installed, allocated, reserved, unavailable and spare switch ports shall reconcile where a governed preliminary port budget is emitted.	Undetermined — engineering input required	Unresolved — engineering input required
AF-NET-SX-001	Network	A Spectrum-X architecture path shall expose	Applicable	Partially supported

Requirement ID	Domain	Requirement	Applicability	Disposition
		candidate-review governance and shall not extrapolate unsupported scale as approved.		
AF-OPS-MODEL-001	Operations	The architecture shall expose customer/provider responsibility boundaries implied by the selected operating model.	Applicable	Unresolved — engineering input required
AF-OPS-OBS-001	Operations	The architecture view set shall expose observability or monitoring capability at Level 2 maturity.	Applicable	Partially supported
AF-SEC-ZONE-001	Security	The architecture view set shall expose security zones or trust boundaries where emitted by the compiler source contract.	Applicable	Partially supported
AF-STOR-FABRIC-001	Storage	The architecture shall declare the selected storage fabric separately from the east-west compute fabric.	Applicable	Unresolved — engineering input required
AF-STOR-GDS-001	Storage	If GPUDirect is requested, the architecture shall expose GPUDirect or RDMA support status and unresolved validation dependencies.	Applicable	Partially supported
AF-STOR-PROTOCOL-001	Storage	Storage protocol requirements shall be explicitly	Applicable	Partially supported

Requirement ID	Domain	Requirement	Applicability	Disposition
		satisfied, diagnosed or assigned to vendor validation.		
AF-STOR-SIZING-001	Storage	Core AI storage outputs shall expose capacity, throughput targets, growth basis and unresolved sizing dependencies.	Applicable	Partially supported
AF-STOR-VENDOR-001	Storage	Where a storage implementation is selected, the architecture shall identify the selected storage pattern, role and, where decided, vendor and governed profile; unresolved selections shall be reported explicitly.	Applicable	Partially supported
AF-TENANCY-001	Security	The architecture shall expose tenancy model and isolation pattern implied by the user request.	Applicable	Unresolved — engineering input required

2.50 Requirements Traceability Matrix

Requirement ID	Architecture Elements	Evidence	Disposition
AF-BOM-SKEL-001	SRC-COMPILER-CONTRACT-W2, ARCH.BOM.ENGINEERING, ARCH.SEMANTIC_TAR GET-fnv1a64-2c2be36cfa35f039, SUP-BOM-SKEL-001, APP-BOM-001, SRC-WEEK4-BCD-CKP	TRACE-fnv1a64-6351c7a0842b69a9, TRACE-fnv1a64-691a3b1383630863, TRACE-fnv1a64-7a612327d4655da9, TRACE-fnv1a64-ca606e07e1e91962, TRACE-fnv1a64-e1ea888c32432383, TRACE-fnv1a64-fb69eac796024e78	Partially supported
AF-COMP-GENERIC-HGX-001	SRC-TRACK-A-SME-SIGNOFF,	TRACE-fnv1a64-029e2898405a9056,	Unresolved — engineering input

Requirement ID	Architecture Elements	Evidence	Disposition
	VAL.GENERIC_PLATFOR M_FACTS, APP-HGX- 001, SUP-COMP- GENERIC-HGX-001	TRACE-fnv1a64- 13a65ac4625c252a, TRACE-fnv1a64- 5f07f5dc254c1543, TRACE-fnv1a64- ec48869d93eec656	required
AF-COMP-PLATFORM- 001	SUP-COMP- PLATFORM-001, APP- HGX-001, ARCH.PLATFORM.SELE CTED, SRC-INPUT- SCHEMA-W2, INPUT- fnv1a64- 3c0ca8ee5d1e9bcc	TRACE-fnv1a64- 362f79b00dd4bdd8, TRACE-fnv1a64- 50060e832987b4cf, TRACE-fnv1a64- 5a0105af08b97bf0, TRACE-fnv1a64- 857ea85fa50d654c, TRACE-fnv1a64- 997239118e97d3db	Unresolved — engineering input required
AF-COMP-SCALE-001	ARCH.MODEL.SCALE, SRC-INPUT-SCHEMA- W2, APP-HGX-001, INPUT-fnv1a64- e1065e064a557b9b, SUP-COMP-SCALE-001	TRACE-fnv1a64- 3ea0d51282568e0a, TRACE-fnv1a64- 7c554b074806ed41, TRACE-fnv1a64- a530e429a1f9c910, TRACE-fnv1a64- c9b7d0d888455665, TRACE-fnv1a64- f5114fee23cdf44a	Unresolved — engineering input required
AF-COMP-SCALE- ROUNDING-001	APP-SCALE- ROUNDING-001, SUP- COMP-SCALE- ROUNDING-001, RAE.STRUCTURED_RE QUIREMENTS_SOURC E, SRC-COMPILER- CONTRACT-W2	TRACE-fnv1a64- 49fd77800b8dad3b, TRACE-fnv1a64- 4e8c5c79f7aabfd8, TRACE-fnv1a64- 619c17faf178659, TRACE-fnv1a64- 77487b77f42b71d2	Partially supported
AF-DD-BLOCKER-001	ARCH.GOVERNANCE. HANDOFF_PACKAGE, APP-ALWAYS-001, SRC-WP3-CONTRACT, SUP-DD-BLOCKER-001	TRACE-fnv1a64- 04929bd28e713cb7, TRACE-fnv1a64- 0b61338249c3e30b, TRACE-fnv1a64- 9a273bb8f086bc61, TRACE-fnv1a64- d346e633380bcc82	Unresolved — engineering input required
AF-EXT-RESP-001	APP-ALWAYS-001, SUP-EXT-RESP-001, SRC-RAE-SPEC-001, RESP-fnv1a64- 50f4762e8580b609, ARCH.GOVERNANCE. HANDOFF_PACKAGE	TRACE-fnv1a64- 0588d64e69cab97c, TRACE-fnv1a64- 10e13508220c5c2a, TRACE-fnv1a64- 9bae914507b8c93d, TRACE-fnv1a64- f363e0ebf5460561,	Partially supported

Requirement ID	Architecture Elements	Evidence	Disposition
AF-FAC-SITE-001	RESP-fnv1a64-0d8d86b3ad012a7f, SUP-FAC-SITE-001, APP-FACILITIES-001, ARCH.SEMANTIC_TAR GET-fnv1a64-dcb9eaddfd9e6b50, SRC-WP3-CONTRACT	TRACE-fnv1a64-fef1f34539b28c19 TRACE-fnv1a64-1e1541d2a51cc0bf, TRACE-fnv1a64-244f3d5cb976c1a2, TRACE-fnv1a64-296e7cc2941a091a, TRACE-fnv1a64-614d5bf6780aa4c1, TRACE-fnv1a64-81e0c53d341c6f8e	Partially supported
AF-GOV-NOLLM-001	SRC-ENGINEERING-PRINCIPLES, COMPILER.DETERMINISTIC_ENGINE, APP-ALWAYS-001, SUP-GOV-NOLLM-001, SRC-RAE-SPEC-001	TRACE-fnv1a64-42f593e250a337ab, TRACE-fnv1a64-538e64da89cc1ba6, TRACE-fnv1a64-9553051b0060963b, TRACE-fnv1a64-98f1c01273266b40, TRACE-fnv1a64-f0045bd8100700e8	Unresolved — engineering input required
AF-GOV-NONPROC-001	SRC-COMPILER-CONTRACT-W2, SUP-GOV-NONPROC-001, SRC-SME-WEEK4-SIGNOFF, VAL.BOM_PROCUREMENT_STATE, APP-ALWAYS-001	TRACE-fnv1a64-08f20673464ff1a9, TRACE-fnv1a64-24e40af815af1442, TRACE-fnv1a64-4511ba418a53730a, TRACE-fnv1a64-8f5cc50c1306462e, TRACE-fnv1a64-cc092a93004e0352	Partially supported
AF-GOV-QSAFETY-001	SUP-GOV-QSAFETY-001, VAL.QUANTITATIVE_SAFETY, SRC-WP3-CONTRACT, APP-ALWAYS-001	TRACE-fnv1a64-470645a4bda7750a, TRACE-fnv1a64-5ea7a78466999c37, TRACE-fnv1a64-6174fb962f4dc2e0, TRACE-fnv1a64-ad06d7be817281c0	Partially supported
AF-GOV-STATUS-001	SRC-PROJECT-STATE-W4, SUP-GOV-STATUS-001, APP-ALWAYS-001, SRC-COMPILER-CONTRACT-W2, ARCH.GOVERNANCE.CORRECTNESS_GATE	TRACE-fnv1a64-128ba3df71d93399, TRACE-fnv1a64-334a60cc6097eaba, TRACE-fnv1a64-4e44f945c2e6bedd, TRACE-fnv1a64-709401d6f7ca09ac, TRACE-fnv1a64-	Unresolved — engineering input required

Requirement ID	Architecture Elements	Evidence	Disposition
AF-GOV-TRACE-001	SUP-GOV-TRACE-001, VAL.TRACE_INTEGRITY, SRC-RAE-SPEC-001, APP-ALWAYS-001	ea3927f627cac339 TRACE-fnv1a64-1134090b5d3e3622, TRACE-fnv1a64-a0d376e7fb683970, TRACE-fnv1a64-b74ae8d2e49dde5d, TRACE-fnv1a64-d4745d05b89f159c	Partially supported
AF-GOV-UNKNOWN-001	SRC-SME-WEEK4-SIGNOFF, VAL.QUANTITATIVE_SAFETY, SRC-WP3-CONTRACT, SUP-GOV-UNKNOWN-001, APP-ALWAYS-001	TRACE-fnv1a64-0004a465af8fce5b, TRACE-fnv1a64-45f37fcd396ae556, TRACE-fnv1a64-5c827e417b8c364d, TRACE-fnv1a64-85527027b5bc1cb8, TRACE-fnv1a64-cd6f1eab7166cc97	Partially supported
AF-NET-EW-001	ARCH.FABRIC.EAST_WEST, INPUT-fnv1a64-378038c20204e1d8, SUP-NET-EW-001, APP-ALWAYS-001, SRC-INPUT-SCHEMA-W2	TRACE-fnv1a64-440502aaf4375619, TRACE-fnv1a64-5e80db04e526cc9e, TRACE-fnv1a64-9d4140c163305422, TRACE-fnv1a64-da677fd3b72f9806, TRACE-fnv1a64-ded11a681d1786a7	Unresolved — engineering input required
AF-NET-IB-001	ARCH.NETWORK.INFINIBAND_TOPOLOGY, SUP-NET-IB-001, APP-FABRIC-IB-001, SRC-IB_TOPOLOGY-CKP	TRACE-fnv1a64-209eb967139f0b8e, TRACE-fnv1a64-30ac09b6c951427e, TRACE-fnv1a64-439d318d30db36ab, TRACE-fnv1a64-7af56dfae63ad0e2	Unresolved — engineering input required
AF-NET-OOB-001	APP-ALWAYS-001, SUP-NET-OOB-001, ARCH.FABRIC.OOB, CAPABILITY.MANAGEMENT_PATH_INDEPENDENCE, SRC-COMPILER-CONTRACT-W2	TRACE-fnv1a64-6827c77b352a3672, TRACE-fnv1a64-93c6cf67896b9b72, TRACE-fnv1a64-adde31eec9d93754, TRACE-fnv1a64-b284f9ffd4117bc3, TRACE-fnv1a64-eeae60020de205c9	Unresolved — engineering input required
AF-NET-PORT-RECON-001	DER-PORT-RECON-001, SRC-TRACK-A-	TRACE-fnv1a64-dd3ab48305207744,	Unresolved — engineering input

Requirement ID	Architecture Elements	Evidence	Disposition
AF-NET-SX-001	SME-SIGNOFF, UNRESOLVED- fnv1a64- 1ae1c1d8cdc46980	TRACE-fnv1a64- e3b361a8c55545e0, TRACE-fnv1a64- f0aff6d593e5eab5	required
	SUP-NET-SX-001, APP- FABRIC-SX-001, ARCH.NETWORK.SPEC TRUMX_FABRIC, SRC- SPECTRUMX-CKP	TRACE-fnv1a64- 0ca98fd33c9c0702, TRACE-fnv1a64- 3c64c76ac7e39d2b, TRACE-fnv1a64- 9f7c5fff7a8e53fa, TRACE-fnv1a64- f85b1146c7496218	Partially supported
AF-OPS-MODEL-001	INPUT-fnv1a64- 75df97b3a025aada, SUP-OPS-MODEL-001, APP-ALWAYS-001, SRC-COMPILER- CONTRACT-W2, ARCH.OPERATIONS.M ODEL	TRACE-fnv1a64- 38307fab308ae19c, TRACE-fnv1a64- 50e2d39eb43c89ea, TRACE-fnv1a64- 636884ed7e338d30, TRACE-fnv1a64- 64372b279e8b088b, TRACE-fnv1a64- ca66209be45a5128	Unresolved — engineering input required
AF-OPS-OBS-001	ARCH.OBSERVABILITY. MODEL, APP- VISUALISATION-001, SRC-VISUALISATION- HS, SUP-OPS-OBS-001	TRACE-fnv1a64- 6ca5d2236d1a66ef, TRACE-fnv1a64- 84013297f558c957, TRACE-fnv1a64- 884b9c6a7b316cd5, TRACE-fnv1a64- bf39be83fa42d2ac	Partially supported
AF-SEC-ZONE-001	ARCH.SECURITY.ZONE S, SUP-SEC-ZONE-001, APP-VISUALISATION- 001, SRC- VISUALISATION-HS	TRACE-fnv1a64- 4b8c08eca6021fa3, TRACE-fnv1a64- 585ccb644842dab2, TRACE-fnv1a64- 7b4f1d02ee99c840, TRACE-fnv1a64- c7ef7d17c3860072	Partially supported
AF-STOR-FABRIC-001	INPUT-fnv1a64- 1046cae5e546f6c4, DER-STORAGE- FABRIC-REQUESTED- 001, DER-STORAGE- FABRIC-SELECTION- 001, ARCH.STORAGE.SELE CTED, SUP-STOR- FABRIC-001, SRC- SPECTRUMX- STORAGE-CKP, APP-	TRACE-fnv1a64- 635510617c17219e, TRACE-fnv1a64- 7fd8510e7e9dace2, TRACE-fnv1a64- 8c0853b067586fa8, TRACE-fnv1a64- a5c3eee3f09949e1, TRACE-fnv1a64- b592c3ccd55ce8ea, TRACE-fnv1a64- c3fb0e284c998211,	Unresolved — engineering input required

Requirement ID	Architecture Elements	Evidence	Disposition
	ALWAYS-001	TRACE-fnv1a64-d6a26f4c8755add0	
AF-STOR-GDS-001	INPUT-fnv1a64-1c284551f886622a, SUP-STOR-GDS-001, APP-GPUDIRECT-001, SRC-STORAGE-CKP, ARCH.SEMANTIC_TAR GET-fnv1a64-85812ade8eb853e1	TRACE-fnv1a64-0774413ea515f935, TRACE-fnv1a64-07e00ef5e9c473fe, TRACE-fnv1a64-37ef31e6d9e69441, TRACE-fnv1a64-4b9e8cf812914952, TRACE-fnv1a64-da58ebaf809c0ca8	Partially supported
AF-STOR-PROTOCOL-001	SRC-STORAGE-CKP, ARCH.SEMANTIC_TAR GET-fnv1a64-f6e123e0c76408f3, APP-STORAGE-VENDOR-001, INPUT-fnv1a64-181424f212208148, SUP-STOR-PROTOCOL-001	TRACE-fnv1a64-09414b9d6b963250, TRACE-fnv1a64-2dd26733a5ff9990, TRACE-fnv1a64-5f7f0c06cccd9392, TRACE-fnv1a64-cdf6f967d00f7715, TRACE-fnv1a64-ee7ec35ac2deddd4	Partially supported
AF-STOR-SIZING-001	SRC-WP3-CONTRACT, ARCH.STORAGE.SIZING_ENVELOPE, APP-STORAGE-VENDOR-001, SRC-VAST-DEFAULT-CKP, SUP-STOR-SIZING-001, INPUT-fnv1a64-1ed6a9a106395cfd, INPUT-fnv1a64-92414a63246c8ed1, INPUT-fnv1a64-53ade9515eac6a5d	TRACE-fnv1a64-26652e65d4027141, TRACE-fnv1a64-27e082d067fe2819, TRACE-fnv1a64-27f68e71040adfa8, TRACE-fnv1a64-69436ec399b8ac91, TRACE-fnv1a64-6fd77157149b5200, TRACE-fnv1a64-bf082edbd393febf, TRACE-fnv1a64-c5052c3736df3ca8, TRACE-fnv1a64-c65924f30f51f8f6	Partially supported
AF-STOR-VENDOR-001	ARCH.STORAGE.SELECTED, SRC-STORAGE-CKP, SUP-STOR-VENDOR-001, ARCH.STORAGE.ARCHITECTURE, APP-STORAGE-VENDOR-001, INPUT-fnv1a64-8a87e8e436180e0b	TRACE-fnv1a64-02d3102d98fb398f, TRACE-fnv1a64-251cb0307f776e92, TRACE-fnv1a64-616ac4e3b03a9298, TRACE-fnv1a64-b0e84d2e8dbdb111, TRACE-fnv1a64-fbbbbc4efffeb68c, TRACE-fnv1a64-fe2903bbd45efdf6	Partially supported

Requirement ID	Architecture Elements	Evidence	Disposition
AF-TENANCY-001	SUP-TENANCY-001, INPUT-fnv1a64- 56c7883ded4941f1, ARCH.TENANCY.MODE L, APP-ALWAYS-001, SRC-COMPILER- CONTRACT-W2	TRACE-fnv1a64- 5384b990c13bcb82, TRACE-fnv1a64- 5895769f6783fa66, TRACE-fnv1a64- 668fb70757e42e40, TRACE-fnv1a64- c6d3f48d2663155c, TRACE-fnv1a64- fbb860805bd92c6f	Unresolved — engineering input required

2.51 Architecture Decision and Approval Record

This section reproduces the human-entered governance records bound to the exact immutable architecture baseline. It does not grant procurement, installation or customer-issue authority.

Record	Identity
Architecture revision	REV-000003
Baseline	BASELINE-000001
Architecture decision	ADR-000001
Authorised review	REVIEW-000001

2.51.1 Architect Decision

- **Selected alternative:** ALT-MAIN
- **Proposed engineering use:** High-level Solution Architecture
- **Rationale:** The preferred design is suitable for bounded solution architecture handoff based on deterministic compiler evidence.
- **Recorded by:** BD Independent SME Reviewer

Trade-offs

- Detailed network design remains required
- Vendor confirmation remains required
- Procurement and installation authority are excluded

2.51.2 Authorised Review

- **Decision:** Approved with conditions
- **Rationale:** Accepted as a bounded Solution Architecture candidate after review of requirements, options, diagrams, BOM and handoff dependencies.
- **Recorded by:** BD Independent SME Reviewer

Approval conditions

- Detailed design validation required
- Vendor sizing and commercial quotation required
- Facilities validation required

2.51.3 Bounded Baseline

- **Classification:** Solution Architecture baseline
- **Approved use:** High-level Solution Architecture

Limitations

- Not procurement ready
- Not installation ready
- Detailed design validation required

2.52 Architecture Baseline Risk Register

This register identifies baseline architectural risks within Neon AI Factory Architect's supported assessment envelope. Compiler findings are advisory inputs to architectural review. They do not constitute customer risk acceptance, implementation validation, vendor certification, or confirmation that proposed controls are effective.

- Baseline findings: 66
- Coverage status: **Incomplete**
- Coverage gaps: 1

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
BARC-COM-001	Compute	Cause: All installed GPU capacity is committed to planned demand. Event: A node is drained, fails or workload demand varies. Consequence: SLA capacity or queue targets cannot be maintained.	Applicable	High; likelihood: Possible; impact: Major	GUI assurance: tenant isolation ownership requires validation before detailed design.	Validate end-to-end capacity
BARC-COM-002	Compute	Cause: Large workloads or services are concentrated on too few schedulable units. Event: A node or rack fails. Consequence: Many jobs restart together and recovery demand	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Spread workloads across failure domains

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
BARC-COM-003	Compute	amplifies the incident. Cause: GPU, NIC, firmware, driver, CUDA, OS and orchestration versions evolve independently. Event: An upgrade or replacement introduces an unsupported combination. Consequence: Performance, stability or vendor support is degraded.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Govern compatibility and lifecycle
BARC-COM-004	Compute	Cause: Spares, access, drain procedures or replacement logistics are unvalidated. Event: A compute component requires repair. Consequence: Reduced capacity persists beyond	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate recovery procedures

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		operational targets.				
BARC-HGB-001	Compute	Cause: The HGX baseboard is integrated with OEM host, power, cooling, NIC and firmware choices. Event: A proposed combination lacks explicit integration evidence. Consequence: Performance, reliability or support expectations may not hold.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Use an approved OEM system profile
BARC-HGB-002	Compute	Cause: GPUs depend on host CPUs, memory, PCIe, firmware and operating system services. Event: A host-side fault or resource bottleneck occurs. Consequence: Multiple GPUs become unavailable	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity; Test under representative failure and load

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		or underutilised.				
BARC-HGB-003	Compute	Cause: Service procedures, spares and firmware tools differ by OEM. Event: A failed component requires cross-vendor diagnosis or replacement. Consequence: Mean time to restore exceeds the architecture assumption.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate recovery procedures
BARC-XDM-001	Cross-domain	Cause: Many interrupted workloads restart, reload data or restore checkpoints together. Event: A rack, node group or control-plane service recovers. Consequence: Storage and fabric saturation prolongs recovery or triggers	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity; Test under representative failure and load

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
BARC-XDM-002	Cross-domain	secondary failures. Cause: Compute and storage compilers use different concurrency , headroom or protocol assumptions. Event: The deployed workload reaches intended GPU scale. Consequence: Storage cannot feed compute at the planned useful throughput.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity
BARC-XDM-003	Cross-domain	Cause: Compute and storage traffic share switches, links, queues or control services. Event: One traffic class saturates or one fabric component fails. Consequence: Both GPU communication and data access degrade together.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Isolate critical traffic domains
BARC-XDM-004	Cross-domain	Cause: Compute,	Applicable	Not yet assessed;	None recorded	Separate correlated

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		storage, network and control components are placed without a common facility-domain model. Event: A power, cooling or rack event occurs. Consequence: Several subsystems lose quorum or capacity simultaneously.		likelihood: Not recorded; impact: Not recorded		dependencies
BARC-XDM-005	Cross-domain	Cause: Recovery actions depend on identity, DNS, automation, OOB or telemetry services with overlapping failures. Event: A data-plane incident also removes a management dependency. Consequence: Operators cannot execute the intended	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies; Validate recovery procedures

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
BARC-XDM-006	Cross-domain	recovery procedure. Cause: Replication volume and recovery windows are not reconciled with edge capacity and failure states. Event: Backlog growth or site recovery requires sustained transfer. Consequence: RPO/RTO objectives cannot be achieved.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity; Test under representative failure and load
BARC-EXT-001	External connectivity	Cause: Edge links are sized independently of workload, replication and tenant peaks. Event: Large transfers or concurrent customer demand occur. Consequence: Datasets, APIs or recovery traffic miss throughput	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		objectives.				
BARC-EXT-002	External connectivity	Cause: Multiple circuits share provider, conduit, meet-me room, router, power or geography. Event: The shared dependency fails. Consequence: External service loses all nominally redundant paths.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies
BARC-EXT-003	External connectivity	Cause: Ownership, prefixes, policy, convergence and route security are not agreed. Event: A path fails or an incorrect route is advertised. Consequence: Connectivity is lost, asymmetric or diverted.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Define and test edge handoff
BARC-EXT-004	External connectivity	Cause: Design, operation, incident and change responsibilities are not	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Define operational ownership

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		assigned at the handoff. Event: A fault or change crosses the boundary. Consequence: Resolution is delayed and unsafe assumptions persist.				
BARC-EXT-005	External connectivity	Cause: Edge connectivity provides broader reach than the customer/s service contract requires. Event: An external identity or system is compromised or misconfigured. Consequence: Internal control, storage or tenant services are exposed.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Harden and restrict access
BARC-IB-001	Compute fabric	Cause: Topology uplinks or rail capacity provide less concurrent bandwidth than endpoint demand.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity; Test under representative failure and load

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		Event: Collective or all-to-all traffic reaches a congested tier. Consequence: Training step time and tail latency increase unpredictably.				
BARC-IB-002	Compute fabric	Cause: Nominally diverse links traverse the same switch, cable route, power zone or manager dependency . Event: The common dependency fails. Consequence: Multiple rails or paths are lost together.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies
BARC-IB-003	Compute fabric	Cause: The selected routing/adaptive-routing policy is not validated for the deployed topology. Event: Traffic becomes imbalanced	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Test under representative failure and load; Implement independent observability

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		or reacts poorly to congestion/failures. Consequence: Links hotspot while available capacity elsewhere is unused.				
BARC-IB-004	Compute fabric	Cause: Discovery, routing and health operations depend on subnet-management services. Event: Management becomes unavailable, split or misconfigured. Consequence: Recovery, topology convergence or operational visibility is impaired.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies; Implement independent observability
BARC-IB-005	Compute fabric	Cause: Large port counts and similar symptoms obscure physical-layer degradation. Event: Intermittent errors or	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Implement independent observability

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		marginal links occur. Consequence: Performance degrades before a clear hard failure identifies the path.				
BARC-SPX-001	Compute fabric	Cause: RDMA performance depends on coordinated NIC, switch, QoS and congestion settings. Event: A hop is misclassified, lossy or inconsistently configured. Consequence: Packet loss, retransmission or pause behaviour degrades AI traffic.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Apply and validate one governed RoCE baseline
BARC-SPX-002	Compute fabric	Cause: Endpoints and switches do not share a validated congestion-control profile. Event: Incast or synchronized AI traffic	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Govern compatibility and lifecycle; Implement independent observability

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		fills queues. Consequence: Tail latency, packet loss or unfairness reduces useful GPU throughput.				
BARC-SPX-003	Compute fabric	Cause: Broad pause domains or incorrect QoS mappings couple unrelated flows. Event: A congested receiver triggers pause propagation . Consequence: Head-of-line blocking affects otherwise healthy traffic.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate bounded QoS and pause domains
BARC-SPX-004	Compute fabric	Cause: Switch, SuperNIC/NIC, firmware and software profiles are version-coupled. Event: A partial lifecycle change departs from the validated RA	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Govern compatibility and lifecycle

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		profile. Consequence: The fabric may operate without intended performance or support characteristics.				
BARC-SPX-005	Compute fabric	Cause: Hashing or adaptive routing does not distribute the actual AI traffic pattern effectively. Event: Large flows collide on a subset of paths. Consequence: Hotspots form while alternate capacity remains idle.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Test under representative failure and load; Implement independent observability
BARC-FAC-001	Facilities	Cause: Aggregate facility power is not reconciled with per-rack feeds, protection and redundancy mode. Event: Peak load or a feed failure transfers demand.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Perform site engineering validation

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		Consequence: Equipment throttles, trips or cannot be commissioned.				
BARC-FAC-002	Facilities	Cause: Flow, pressure, temperature, water quality or heat rejection is not validated end to end. Event: Facility cooling deviates under load or failure. Consequence: Compute throttles, alarms, shuts down or suffers damage risk.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate liquid-cooling chain
BARC-FAC-003	Facilities	Cause: High-density racks and distribution hardware exceed assumed structural or handling limits. Event: Installation, movement or operation	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate structure and logistics

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		applies excessive load. Consequence: Deployment is delayed or structural/handling safety is compromised.				
BARC-FAC-004	Facilities	Cause: Power, cooling or fire-protection maintenance domains are broader than compute scheduling domains. Event: Planned facility maintenance occurs. Consequence: More AI capacity is unavailable than the service plan anticipates.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies; Validate recovery procedures
BARC-FAC-005	Facilities	Cause: Sensors and alarms do not cover rack inlet, coolant, leaks, power quality and local hotspots. Event: A local	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Implement independent observability

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		environmental deviation develops. Consequence: Operators detect it after throttling or equipment alarms.				
BARC-OOB-001	Management	Cause: Management and production paths share switches, power, routing or authentication. Event: A production incident disrupts the shared dependency. Consequence: Operators lose the recovery path when it is most needed.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies
BARC-OOB-002	Management	Cause: Powerful management interfaces are reachable by excessive identities or networks. Event: Credentials or a management endpoint	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Harden and restrict access

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		are compromised. Consequence: An attacker can disrupt or reconfigure infrastructure at scale.				
BARC-OOB-003	Management	Cause: Dual OOB paths share cabling routes, upstream gateways or facility domains. Event: The common dependency fails. Consequence: Remote administration becomes unavailable.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies; Test under representative failure and load
BARC-OOB-004	Management	Cause: Emergency identities, consoles or local procedures are documented but not exercised. Event: Normal identity or network services are unavailable. Consequence: Recovery is delayed or impossible.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate recovery procedures

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
BARC-SRV-001	Service network	Cause: Multiple trust zones or tenants share broad network reachability. Event: A compromised or misconfigured workload reaches another service boundary. Consequence: Confidentiality, integrity or availability is affected across tenants.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Harden and restrict access
BARC-SRV-002	Service network	Cause: Ingress, egress, API, data and management flows share insufficient capacity. Event: Peak customer or data-transfer demand occurs. Consequence: Customer-visible latency or throughput breaches objectives.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity
BARC-SRV-	Service	Cause:	Applicable	Not yet	None	Separate

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
003	network	DNS, DHCP, load balancing, routing or identity dependencies lack independent availability. Event: A shared control service fails. Consequence: Healthy compute services become unreachable or unable to recover.		assessed; likelihood: Not recorded; impact: Not recorded	recorded	correlated dependencies; Validate recovery procedures
BARC-SRV-004	Service network	Cause: Telemetry cannot attribute congestion, drops or policy violations to tenant and path. Event: A noisy-neighbour or fault condition occurs. Consequence: Diagnosis and fair remediation are delayed.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Implement independent observability
BARC-OPS-001	Operations	Cause: Workload placement, recovery and lifecycle depend on	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies; Validate recovery procedures

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		concentrated control services. Event: The control plane is unavailable or inconsistent. Consequence: Healthy compute cannot accept, recover or manage workloads.				
BARC-OPS-002	Operations	Cause: Deployed firmware, network, storage or platform state diverges from the approved baseline. Event: An incident or performance issue occurs on an unknown configuration. Consequence: Supportability and diagnosis are impaired.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Govern compatibility and lifecycle; Implement independent observability
BARC-OPS-003	Operations	Cause: Compute, fabric, storage and orchestration	Applicable	Not yet assessed; likelihood: Not recorded;	None recorded	Govern compatibility and lifecycle

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		n changes are planned independently. Event: A staged upgrade creates a transient unsupported combination. Consequence: Service instability or rollback complexity increases.		impact: Not recorded		
BARC-OPS-004	Operations	Cause: Runbooks focus on single-component replacement and normal management availability. Event: A wider domain or management outage occurs. Consequence: Operators improvise under pressure and recovery exceeds objectives.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate recovery procedures; Test under representative failure and load
BARC-OPS-005	Operations	Cause: Metrics cover availability but not utilisation,	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not	None recorded	Implement independent observability

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		errors, queues, tail latency and cross-domain correlation. Event: A soft failure or resource contention develops. Consequence: GPU productivity falls before operators isolate the cause.		recorded		
BARC-OPT-001	Optical connectivity	Cause: Transceiver, fibre type, connector, breakout and reach are not validated as one channel. Event: The installed channel exceeds loss/reach limits or contains incompatible parts. Consequence: Links fail, flap or operate with reduced margin.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate the optical channel
BARC-OPT-002	Optical connectivity	Cause: Redundant logical links share trays, trunks, patch	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not	None recorded	Separate correlated dependencies

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		panels or routes. Event: A cable or pathway incident occurs. Consequence: Several intended redundant links fail together.		recorded		
BARC-OPT-003	Optical connectivity	Cause: Dense cabling lacks authoritative labels, mapping, spares or polarity records. Event: A link is installed, moved or replaced. Consequence: Mismatching or slow fault isolation extends outage time.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate recovery procedures; Implement independent observability
BARC-SEC-001	Security	Cause: Assets or users gain broad access because they are inside a network boundary. Event: An internal endpoint or identity is compromised	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Harden and restrict access

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		ed. Consequence: The attacker moves laterally to high-value AI infrastructure.				
BARC-SEC-002	Security	Cause: Shared or highly privileged accounts span compute, network, storage and facilities. Event: A credential is misused or compromised. Consequence: Multiple architecture domains can be altered without independent control.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Harden and restrict access
BARC-SEC-003	Security	Cause: Credentials, tokens or keys are embedded in code, images, logs or configuration exports. Event: An artefact is copied or accessed outside its	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Use governed secrets management

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		intended boundary. Consequence: Infrastructure services are accessed without authorisation.				
BARC-SEC-004	Security	Cause: Administrative, identity, network and workload events are incomplete, unsynchronised or mutable. Event: A security or operational incident occurs. Consequence: Scope, sequence and accountability cannot be established promptly.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Implement independent observability
BARC-SEC-005	Security	Cause: Firmware, images, packages or configurations lack verified origin and approved baselines. Event: A vulnerable, tampered or unintended component	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Govern compatibility and lifecycle

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		is introduced. Consequence: Security and support assumptions no longer hold.				
BARC-STO-001	Storage	Cause: Usable demand is compared directly with raw capacity. Event: Protection, rebuild, metadata, snapshots or growth consume unplanned capacity. Consequence: The cluster reaches operational limits earlier than planned.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity
BARC-STO-002	Storage	Cause: Sizing uses aggregate bandwidth without IOPS, metadata, concurrency or read/write mix. Event: Checkpoint, ingest, shuffle or small-file phases dominate.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Test under representative failure and load

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		Consequence: GPUs stall despite acceptable peak storage bandwidth.				
BARC-STO-003	Storage	Cause: Local redundancy is treated as protection from deletion, corruption or site loss. Event: A logical or site-level incident affects all live copies. Consequence: Required data cannot be restored.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Provide independent tested recovery copies
BARC-STO-004	Storage	Cause: Logical node counts are not mapped to rack, power, cooling and switch domains. Event: One physical event removes more components than protection tolerates. Consequence: Availability or data-protection	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		objectives may not be met.				
BARC-WEK-001	Storage	Cause: Configured failure domains differ from actual shared power, rack or network dependencies. Event: A common cause removes multiple declared domains. Consequence: The protection tolerance is exceeded.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies
BARC-WEK-002	Storage	Cause: DPDK, UDP, RDMA, SR-IOV or CPU allocations are heterogeneous or under-provisioned. Event: Traffic uses a degraded path or node-local network bottleneck. Consequence: Performance or availability becomes uneven across the	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Govern compatibility and lifecycle; Test under representative failure and load

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
BARC-WEK-003	Storage	cluster. Cause: Expansion adds unequal failure-domain capacity or materially different media. Event: Rebalancing and steady-state placement encounter heterogeneity. Consequence: Capacity efficiency, performance or operational predictability is reduced.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Use a governed expansion plan
BARC-WEK-004	Storage	Cause: Protection level, hot spare/reserve and usable capacity are selected independently of outage requirements. Event: Concurrent failures occur during rebuild or high utilisation. Consequence: Recovery	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity; Implement independent observability

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		margin or capacity availability becomes insufficient.				
BARC-SYS-001	System-wide	Cause: A required service path contains an unprotected dependency . Event: That dependency fails or is unavailable during maintenance. Consequence: The AI service becomes unavailable despite redundancy elsewhere.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies; Test under representative failure and load
BARC-SYS-002	System-wide	Cause: Compute, network, storage and facilities use different or undocumented failure-domain mappings. Event: One physical event crosses several logical protection groups. Consequence: A correlated outage exceeds the	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Separate correlated dependencies

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		tolerance assumed by individual subsystems .				
BARC-SYS-003	System-wide	Cause: Subsystems are sized independently without reconciling peak demand and headroom. Event: Workload demand reaches the weakest compute, fabric, storage or edge constraint. Consequence: Delivered AI throughput or service quality falls below the architecture objective.	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Validate end-to-end capacity
BARC-SYS-004	System-wide	Cause: Material design choices depend on assumptions not represented as validated facts. Event: A customer, site or workload condition contradicts an	Applicable	Not yet assessed; likelihood: Not recorded; impact: Not recorded	None recorded	Govern assumptions

Risk ID	Category	Baseline risk	Applicability	Human assessment	Review note	Candidate treatments
		assumption. Consequence: Sizing, resilience or supportability conclusions require redesign.				

2.53 Evidence Appendix

2.54 Vendor Options

The Vendor Options Compiler presents the governed implementation-option view for this exact architecture revision.

2.54.1 Executive Vendor Options Summary

Architecture area	Current architecture	Compatible implementations	Alternative architectures	Conclusion
Compute	NVIDIA GB300 NVL72	None established	None represented	The governed catalogue currently represents the selected architecture basis only.
Network	Architecture requirement represented	NVIDIA Spectrum-4 SN5600	Arista 7060X6-64PE, Cisco N9364E-SG2-O, NVIDIA Quantum-X800 / Q3400 family	Governed candidates are represented; further technical evidence is required for some options.
Optical Connectivity	Architecture requirement represented	None established	None represented	Governed candidates are represented; further technical evidence is required for some options.
Storage	VAST Data Platform	None established	DDN AI/HPC storage platform, WEKA Data Platform / WEKApod family	The current basis is retained and alternative architecture options are visible for review.

2.54.2 Detailed Requirement Comparison

2.54.2.1 Rack-scale GPU compute system

- Architecture area: Compute
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required

- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **NVIDIA GB300 NVL72** — Selected architecture basis; Current requirement quantity applies

2.54.2.2 *AI compute-fabric leaf switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Arista 7060X6-64PE** — Alternative architecture option; Independent architecture-branch sizing required
- **Cisco N9364E-SG2-O** — Alternative architecture option; Independent architecture-branch sizing required
- **NVIDIA Quantum-X800 / Q3400 family** — Alternative architecture option; Independent architecture-branch sizing required
- **NVIDIA Spectrum-4 SN5600** — Compatible implementation option; Current requirement quantity applies

2.54.2.3 *AI compute-fabric spine switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Arista 7060X6-64PE** — Alternative architecture option; Independent architecture-branch sizing required
- **Cisco N9364E-SG2-O** — Alternative architecture option; Independent architecture-branch sizing required
- **NVIDIA Quantum-X800 / Q3400 family** — Alternative architecture option; Independent architecture-branch sizing required
- **NVIDIA Spectrum-4 SN5600** — Compatible implementation option; Current requirement quantity applies

2.54.2.4 *Storage-fabric leaf switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Arista 7060X6-64PE** — Further technical evidence required; Quantity applicability not established
- **Cisco N9364E-SG2-O** — Further technical evidence required; Quantity applicability not established
- **NVIDIA Spectrum-4 SN5600** — Further technical evidence required; Quantity applicability not established

2.54.2.5 *Storage-fabric spine switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Arista 7060X6-64PE** — Further technical evidence required; Quantity applicability not established
- **Cisco N9364E-SG2-O** — Further technical evidence required; Quantity applicability not established
- **NVIDIA Spectrum-4 SN5600** — Further technical evidence required; Quantity applicability not established

2.54.2.6 *Service/tenant leaf switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Cisco Nexus 93180YC-FX3** — Further technical evidence required; Quantity applicability not established
- **Dell PowerSwitch S5248F-ON** — Further technical evidence required; Quantity applicability not established
- **Juniper QFX5120-48Y** — Further technical evidence required; Quantity applicability not established

2.54.2.7 *Service/tenant spine switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.

2.54.2.8 *Out-of-band access switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Arista CCS-720XP-48Y6** — Further technical evidence required; Quantity applicability not established
- **Cisco Catalyst C9300-48T** — Further technical evidence required; Quantity applicability not established
- **NVIDIA SN2201** — Further technical evidence required; Quantity applicability not established

2.54.2.9 *Out-of-band aggregation switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Arista CCS-720XP-48Y6** — Further technical evidence required; Quantity applicability not established
- **Cisco Catalyst C9300-48T** — Further technical evidence required; Quantity applicability not established
- **Cisco Nexus 93180YC-FX3** — Further technical evidence required; Quantity applicability not established
- **Dell PowerSwitch S5248F-ON** — Further technical evidence required; Quantity applicability not established
- **Juniper QFX5120-48Y** — Further technical evidence required; Quantity applicability not established
- **NVIDIA SN2201** — Further technical evidence required; Quantity applicability not established

2.54.2.10 *Out-of-band core switch*

- Architecture area: Network
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Cisco Nexus 93180YC-FX3** — Further technical evidence required; Quantity applicability not established
- **Dell PowerSwitch S5248F-ON** — Further technical evidence required; Quantity applicability not established
- **Juniper QFX5120-48Y** — Further technical evidence required; Quantity applicability not established

2.54.2.11 *800G optical transceiver*

- Architecture area: Optical Connectivity
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Arista OSFP-800G-2FR4** — Further technical evidence required; Quantity applicability not established
- **Arista OSFP-800G-2XDR4** — Further technical evidence required; Quantity applicability not established
- **Cisco OSFP-800G-DR8P** — Further technical evidence required; Quantity applicability not established

- **NVIDIA 2x400GbE twin-port OSFP 100m single-mode transceiver 980-9I30H-F4NM00** — Further technical evidence required; Quantity applicability not established
- **NVIDIA 800G twin-port OSFP DR8/MMF transceiver 980-9I510-F4NS00** — Further technical evidence required; Quantity applicability not established

2.54.2.12 *Core AI storage platform*

- Architecture area: Storage
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **DDN DDN AI/HPC storage platform** — Alternative architecture option; Independent architecture-branch sizing required
- **VAST Data VAST Data Platform** — Selected architecture basis; Current requirement quantity applies
- **WEKA WEKA Data Platform / WEKApod family** — Alternative architecture option; Independent architecture-branch sizing required

2.54.2.13 *Storage controller/frontend node*

- Architecture area: Storage
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **VAST Data Gen6 CBox** — Selected architecture basis; Current requirement quantity applies

2.54.2.14 *Storage capacity node*

- Architecture area: Storage
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **VAST Data Ceres v2 DBox** — Selected architecture basis; Current requirement quantity applies

2.54.2.15 *Storage management node*

- Architecture area: Storage
- Requirement: Architecture detail not yet available
- Quantity: Unresolved - engineering input required
- Coverage: The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **VAST Data VMS node** — Selected architecture basis; Current requirement quantity applies

2.54.3 Unresolved Evidence and Sizing

- **Rack-scale GPU compute system:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **AI compute-fabric leaf switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **AI compute-fabric spine switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Storage-fabric leaf switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Storage-fabric spine switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Service/tenant leaf switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Service/tenant spine switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Out-of-band access switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Out-of-band aggregation switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Out-of-band core switch:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **800G optical transceiver:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Core AI storage platform:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Storage controller/frontend node:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Storage capacity node:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.
- **Storage management node:** Architecture detail not yet available. The governed acquisition class is retained, but the accepted architecture does not yet provide the required typed detail.

Vendor Options identifies governed compatible implementations and architecture alternatives based on the knowledge available for the selected architecture revision. Some alternatives may require further technical validation or independent branch sizing. Inclusion does not constitute procurement approval, commercial recommendation, availability confirmation, or vendor commitment.

2.55 Phased Physical Realization

This section is calculated from the accepted architecture intent and a governed allocation profile. It provides engineering review evidence, not procurement or installation authority.

The RAC-selected reference fabric supports **144 systems / 1,152 GPUs**. The phased deployment plans an active target population of **128 systems / 1,024 GPUs**, leaving 16 supported node positions unpopulated. Installed fabric capacity is not presented as active GPU population.

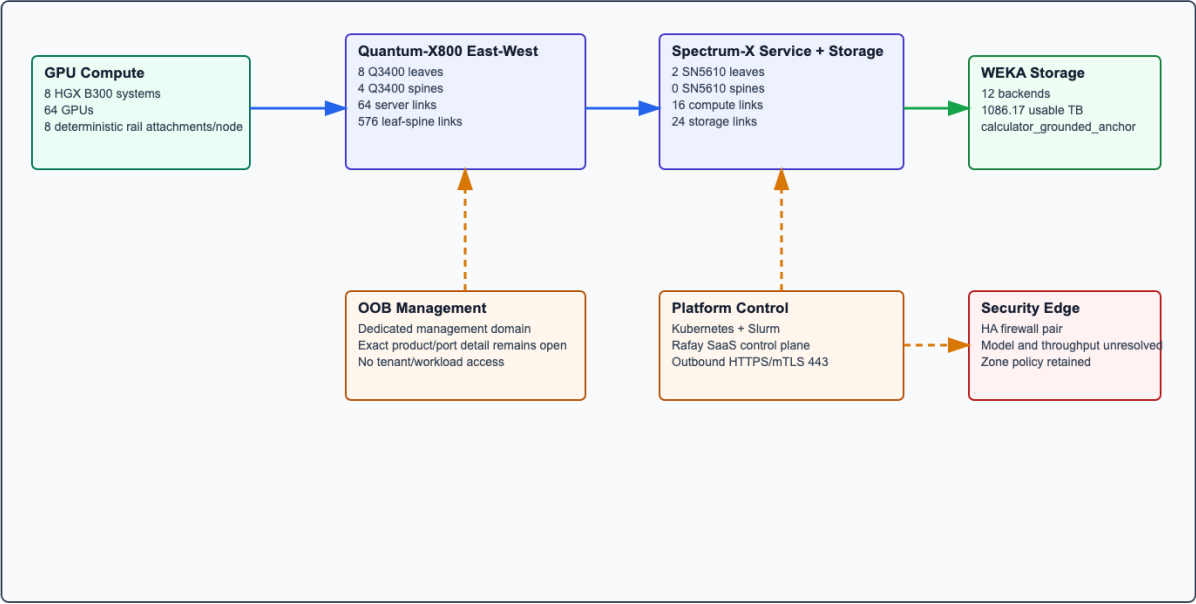
Phase	Compute systems	GPUs	WEKA backends	Quantum-X800 leaf/spine	Spectrum-X leaf/spine	WEKA usable capacity
Foundation	8	64	12	8 / 4	2 / 0	1086.17 TB
Expansion	16	128	24	8 / 4	2 / 4	2172.34 TB
Target	128	1024	48	16 / 8	8 / 4	4344.68 TB

2.55.1 Overall Phased AI Factory Physical Realization

680 physical ledger links reconcile to this grouped view. SVG digest:
sha256:bb1efc6bcd036e3b2d01e0cf65c719efa50302ca817064d2d214c17fe80cd8bb8.

Overall Phased AI Factory Physical Realization

Foundation | 64 GPUs | exact ledger references retained



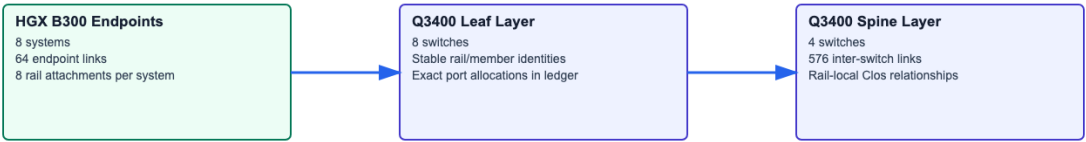
Overall Phased AI Factory Physical Realization

2.55.2 Quantum-X800 Physical Fabric

640 physical ledger links reconcile to this grouped view. SVG digest:
sha256:0a0b810c3f27603adfeabe612790c294dc48f446a7871ca3f8683ad2d730d30b.

Quantum-X800 Physical Fabric

Foundation | 64 GPUs | exact ledger references retained

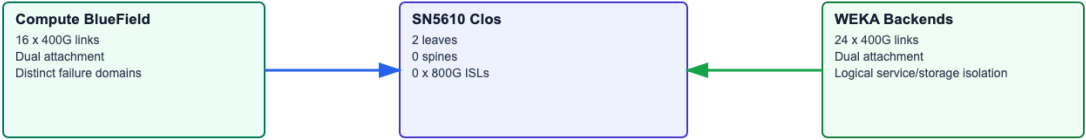


Quantum-X800 Physical Fabric

2.55.3 Spectrum-X Service and Storage Fabric

40 physical ledger links reconcile to this grouped view. SVG digest:
sha256:089ce7ffe351d8cb9d69ea60d8502516ed13e29c254ec9269b6a2fe7340a3dff.

Spectrum-X Service and Storage Fabric
Foundation | 64 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

Spectrum-X Service and Storage Fabric

2.55.4 WEKA Capacity and Attachment

24 physical ledger links reconcile to this grouped view. SVG digest:
sha256:7a324c7a494b3522ddd2b37ce0beebf90136e759a2be4d0ba007ea6b95600706.

WEKA Capacity and Attachment

Foundation | 64 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

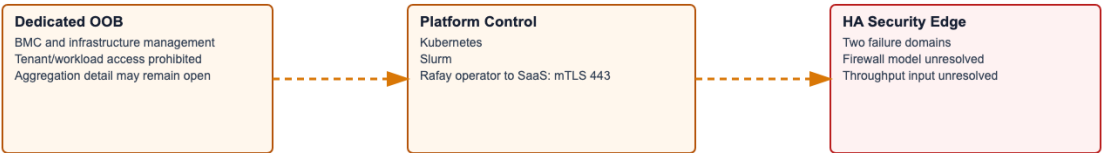
WEKA Capacity and Attachment

2.55.5 Management, Platform and Security Boundaries

0 physical ledger links reconcile to this grouped view. SVG digest:
sha256:d3458d59242e5b6622450950fe78f0f1ec48c8dd0882b2513e45b8f30575fb99.

Management, Platform and Security Boundaries

Foundation | 64 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

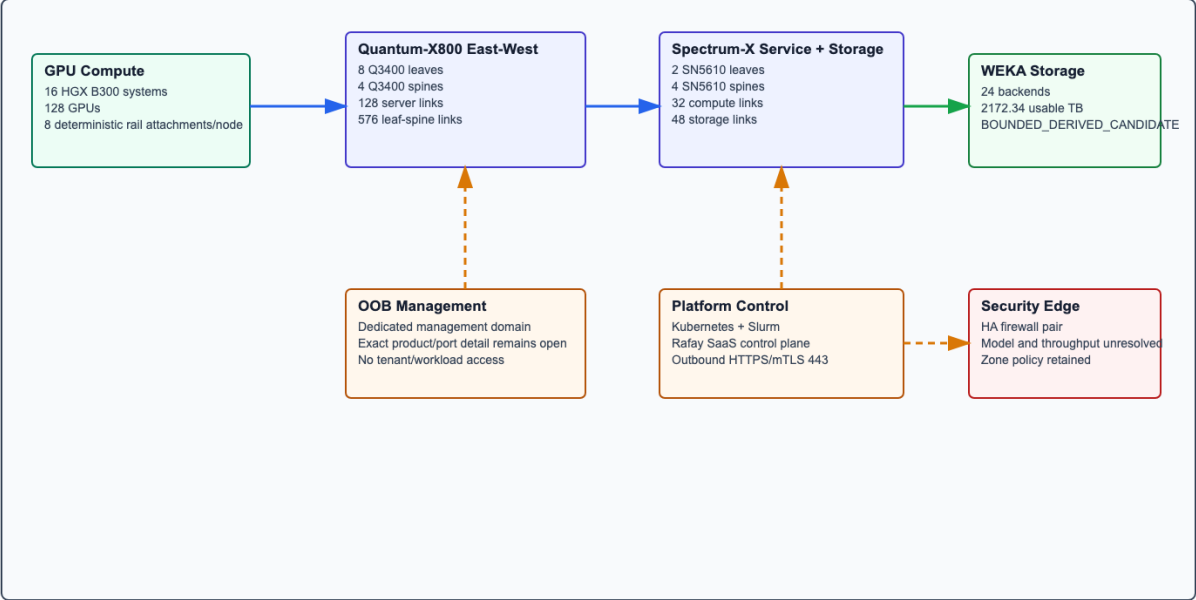
Management, Platform and Security Boundaries

2.55.6 Overall Phased AI Factory Physical Realization

848 physical ledger links reconcile to this grouped view. SVG digest:
sha256:23bc0467fa3a79b7d80b28fddff18420ea9687cd9dddf017a49306861d0c93f5.

Overall Phased AI Factory Physical Realization

Expansion | 128 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

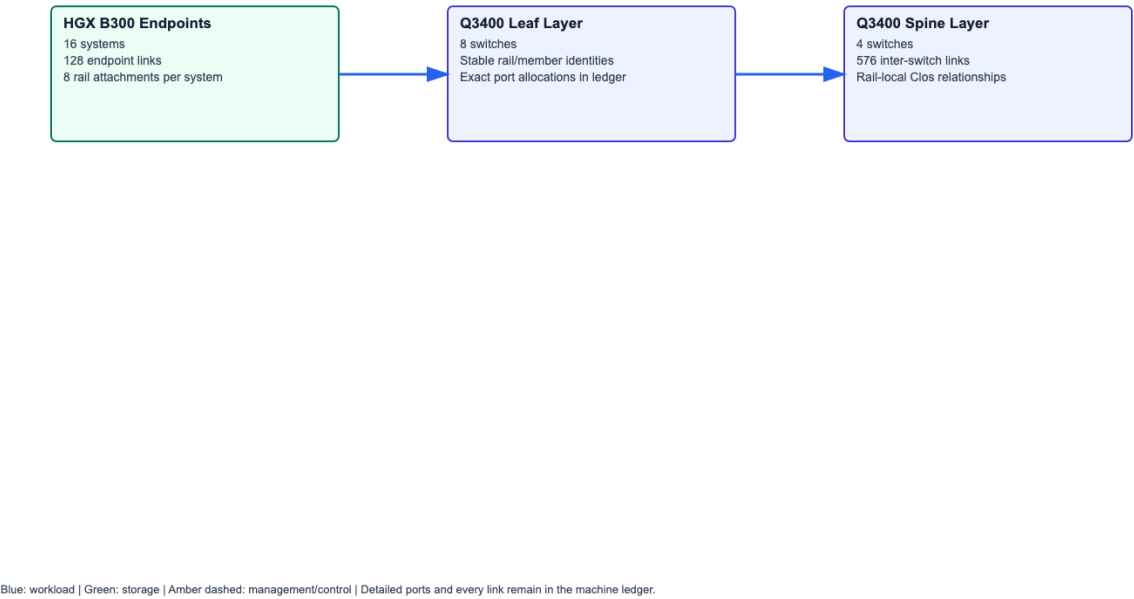
Overall Phased AI Factory Physical Realization

2.55.7 Quantum-X800 Physical Fabric

704 physical ledger links reconcile to this grouped view. SVG digest:
sha256:77f55cec4dd3f77d92f80c41cc9be6127204be0654ffab94a6c3053658f98603.

Quantum-X800 Physical Fabric

Expansion | 128 GPUs | exact ledger references retained



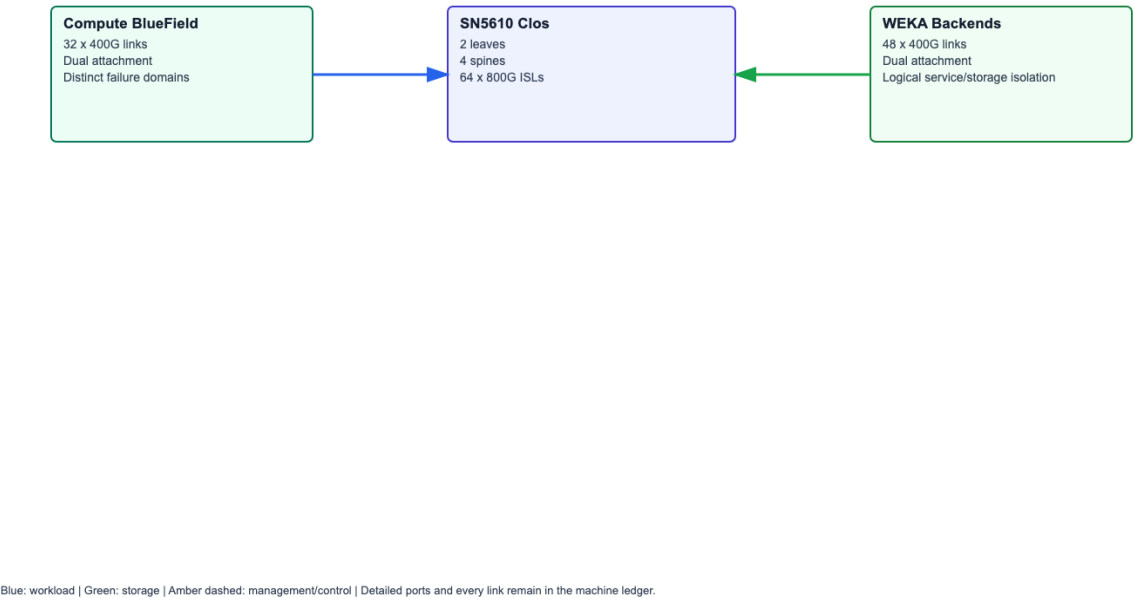
Quantum-X800 Physical Fabric

2.55.8 Spectrum-X Service and Storage Fabric

144 physical ledger links reconcile to this grouped view. SVG digest:
sha256:80e6558d7dd7187007f49abf100b6d18fc5f1cd817bc4c8061b395205b93d325.

Spectrum-X Service and Storage Fabric

Expansion | 128 GPUs | exact ledger references retained



Spectrum-X Service and Storage Fabric

2.55.9 WEKA Capacity and Attachment

48 physical ledger links reconcile to this grouped view. SVG digest:
sha256:37255d7de1773199ccbe7a2336123f8295db07180b1374242b8670b226f68d15.

WEKA Capacity and Attachment
Expansion | 128 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

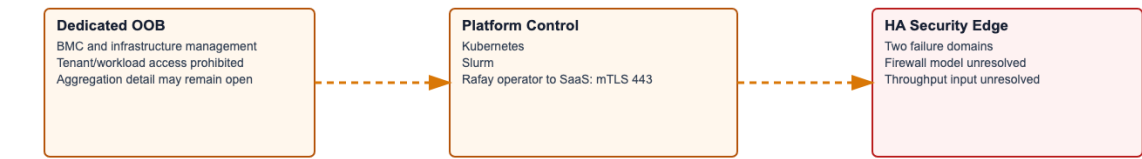
WEKA Capacity and Attachment

2.55.10 Management, Platform and Security Boundaries

0 physical ledger links reconcile to this grouped view. SVG digest:
sha256:599f46e53e73d5e79b9424e37b24c366cd8ce93dffa407dc9e41c506b210a1ff.

Management, Platform and Security Boundaries

Expansion | 128 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

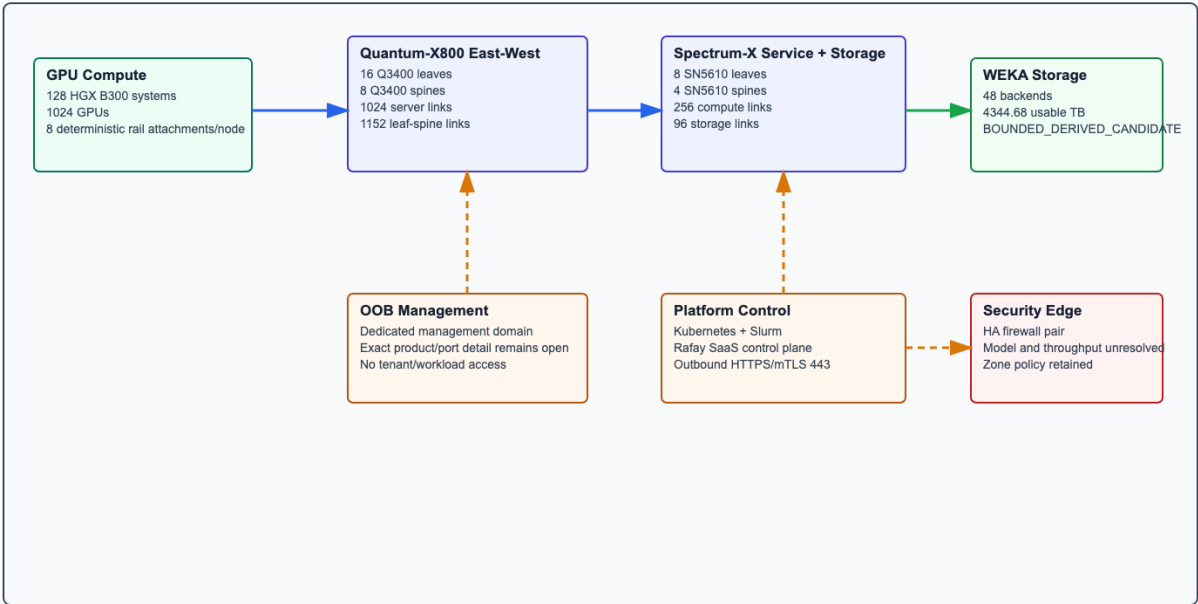
Management, Platform and Security Boundaries

2.55.11 Overall Phased AI Factory Physical Realization

2784 physical ledger links reconcile to this grouped view. SVG digest:
sha256:e511ed58c07f0ce5fa952f2c6d4f5f65218a23d1e69041c6051cbac4c91586b2.

Overall Phased AI Factory Physical Realization

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

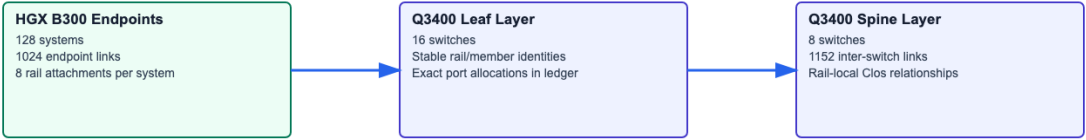
Overall Phased AI Factory Physical Realization

2.55.12 Quantum-X800 Physical Fabric

2176 physical ledger links reconcile to this grouped view. SVG digest:
sha256:bf23d285f17d6dbf1cfaa62536eac3aff23a39c0e531c433af6780e0ff55ce99.

Quantum-X800 Physical Fabric

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

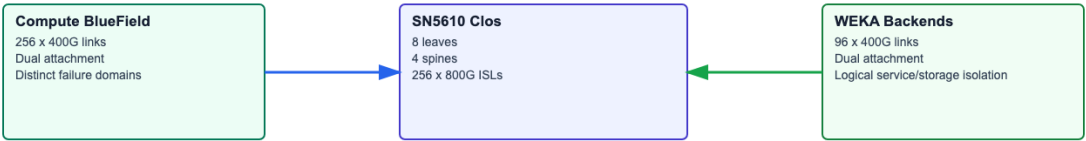
Quantum-X800 Physical Fabric

2.55.13 Spectrum-X Service and Storage Fabric

608 physical ledger links reconcile to this grouped view. SVG digest:
sha256:0f41826470ae579cba8f01c3eeb143e670dd8a87ca63c2a671243bcf53eaa5b4.

Spectrum-X Service and Storage Fabric

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

Spectrum-X Service and Storage Fabric

2.55.14 WEKA Capacity and Attachment

96 physical ledger links reconcile to this grouped view. SVG digest:
sha256:092b8e4174415bb99f5d9e1640e3bb592ca66f2474857ded942af8009b612c7a.

WEKA Capacity and Attachment

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

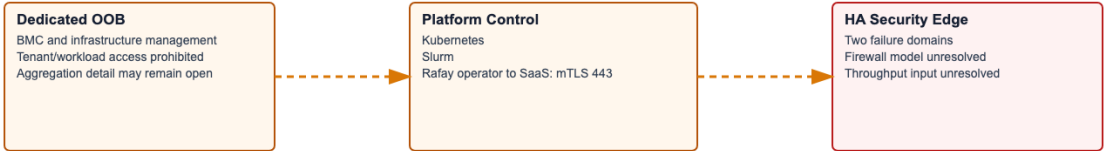
WEKA Capacity and Attachment

2.55.15 Management, Platform and Security Boundaries

0 physical ledger links reconcile to this grouped view. SVG digest:
sha256:4edbe184a1546a921e0bbfb1de5443afc6682179b182793e196ff5fffb9c55142.

Management, Platform and Security Boundaries

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

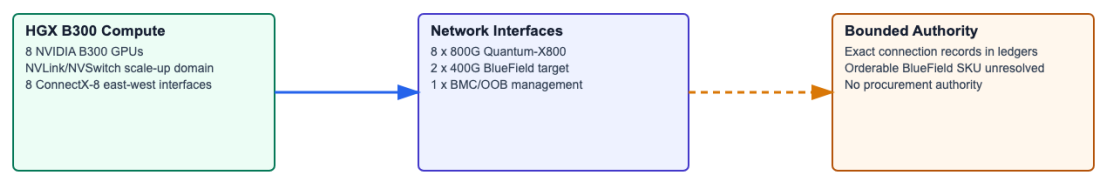
Management, Platform and Security Boundaries

2.55.16 HGX B300 Node Architecture

1408 physical ledger links reconcile to this grouped view. SVG digest:
sha256:3753ff6792a6e0c5f2626aa3ef93cd12e20f02a539f972176f50ef5d12508304.

HGX B300 Node Architecture

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

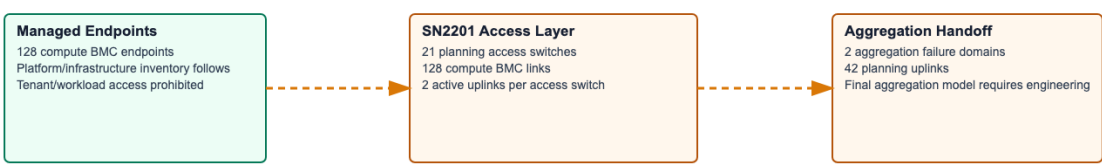
HGX B300 Node Architecture

2.55.17 OOB Management Topology

170 physical ledger links reconcile to this grouped view. SVG digest:
sha256:22fc4cdef5cfdab57cef0b0d5ddf9ffd95a598bb56e083d2538894ee1010e76a.

OOB Management Topology

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

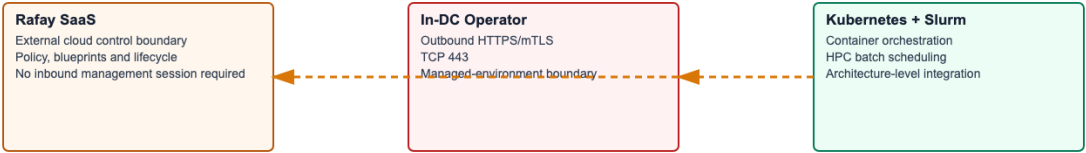
OOB Management Topology

2.55.18 Rafay Enterprise Control Architecture

0 physical ledger links reconcile to this grouped view. SVG digest:
sha256:19e405809980d448756bdcfd4959d79972ff07c60e7a6afe619a7a8728e006fc.

Rafay Enterprise Control Architecture

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

Rafay Enterprise Control Architecture

2.55.19 Enterprise Software Stack

0 physical ledger links reconcile to this grouped view. SVG digest:
sha256:5f50a2109bec471c8643b32770d853af1fc8b680aac2cc22983a2ed86e42d00a.

Enterprise Software Stack

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

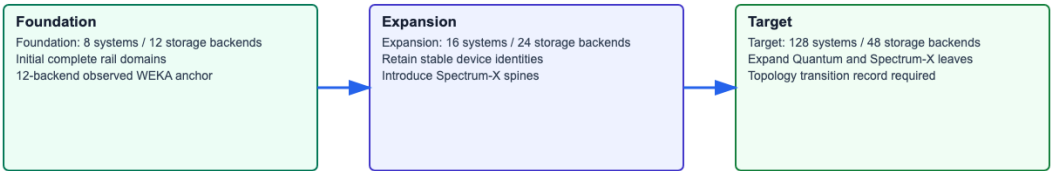
Enterprise Software Stack

2.55.20 Phased Architecture Evolution

0 physical ledger links reconcile to this grouped view. SVG digest:
sha256:6edf23d4e1af8f1e92f3960a9571fece9313a3c6d94522329048ee983b8acbee.

Phased Architecture Evolution

Target | 1024 GPUs | exact ledger references retained



Blue: workload | Green: storage | Amber dashed: management/control | Detailed ports and every link remain in the machine ledger.

Phased Architecture Evolution

The engineering evidence set contains **143 deterministic vector drill-down views** with a canonical link/device/port index

(sha256:828ce49182ce3b6975885b63699cdff9910d4e34ebd189908544e12b2c878f88).

These supporting views retain exact physical traceability without rasterizing every link sheet into this document.

Stable device identities and topology-transition records are retained across phases. WEKA Phase 1 preserves the production-observed consultant-confirmed evidence state; Phase 2 and Phase 3 remain bounded derived candidates pending vendor validation.

2.56 No-Loss Reconciliation

- Total facts: 24
- Covered facts: 24
- Complete: Yes

2.57 ADC v1 Quality Gates

- Gate set: Professional SADM v1
- Professional gate status: **Passed**
- Blocking failures: 0
- Non-blocking failures: 1

2.58 ADC v1 Figure Resolution

- Main visualisations: 1
- Conceptual topology figures: 33
- Materialised figures: 52
- Source-only figures requiring materialisation: 0